

Mathematical introduction

Branching, quasi-stationarity, and absorption models

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Chapter 2

Mathematical introduction

2.1 Overview

This chapter introduces the mathematical framework and the methods used throughout the thesis. Its concern is the machinery rather than any single application: the process classes in which the models live, and the stock of exact arguments by which those processes are analysed. Later chapters build their models from the objects defined here, and appeal to the methods developed here; the present one aims to make both available in a form that can be cited rather than re-derived.

Two questions organise everything below, and it is worth naming them because they are not the same question. The first is a *distributional* question: given the process, what is the law of the population — its mean, its variance, its full generating function — at a given time, or in the limit? The second is an *absorption* question: does the process reach a given state at all, and if so with what probability, or after what time? Extinction, rupture and establishment are all absorption questions; growth and quasi-stationarity are distributional ones. The methods of [section 2.3](#) split accordingly. First-step analysis and its continuous-time companions answer absorption questions; generating functions and moment equations answer distributional ones; and where a problem mixes the two — survival *conditioned on* not being absorbed, for instance — the two stocks of methods are used together, as in [section 2.3.2.6](#).

[Section 2.2](#) defines the process classes. Discrete-time Markov chains are introduced through simple random walks and Galton–Watson processes; the latter is the ancestor of every branching model in the thesis, and is treated here only as far as its definition and one-step structure, with the critical theory in [section 2.C](#) and the deep analysis of its quasi-stationary amplitude in Chapter 3 (*The constant $A(p)$*). Continuous-time chains follow, through the Poisson process and the birth–death and birth–death–catastrophe processes, the latter carrying the second absorbing mechanism — abrupt rupture — that several later chapters need. Time-inhomogeneous chains, whose rates vary with the clock, are set up with the logistic speciation model as the worked instance; and coupled ODE–CTMC systems, in which a deterministic flow and a jump process drive one another, close the section with the rupture-into-a-medium schematic on which the compartment chapters are built.

[Section 2.3](#) then develops the methods, again in two time domains. The discrete-time stock — generating functions, functional iteration, first-step analysis — is set out briefly, because its deeper development belongs to Chapter 3 (*The constant $A(p)$*), where the binary Galton–Watson process is analysed by Koenigs-function methods. The continuous-time stock is the main technical weight of the chapter: backward equations, mean population size, variance and higher moments, hitting probabilities, extinction probabilities, and conditional means, all developed with the birth–death

process as the running example. The conditional-mean subsection is where the two halves of the thesis meet: it defines the discrete constant $A(p)$ and the quasi-stationary mean $1/A(p)$, derives the elementary continuous-time constant $A_c(p)$ in full, and hands the analysis of $A(p)$ itself — product and series representations, bounds, near-critical asymptotic, closed-form searches, and the dynamical obstruction to any closed form — to Chapter 3 (*The constant $A(p)$*).

One method crosses the division and receives its own treatment in [section 2.3.3](#): the method of characteristics, by which the first-order partial differential equations satisfied by generating functions are solved. It is presented as a four-step recipe and worked through on the absorption–death model; the harder members of the family, including the absorption–birth–death model whose closed form is an original result of this thesis, are solved in [section 2.G](#) by the identical procedure, supported by the integral identity of [section 2.A](#) and the coefficient-extraction machinery of [section 2.B](#). The remaining appendices hold material displaced from the main text by the chapter’s framework-and-methods organisation — the critical theory of branching, the arithmetic of founding cohorts, the quasi-stationary variance, and the discrete catastrophe calculation — so that nothing the later chapters need is lost.

Two conventions run throughout. Processes are specified by their rates and nothing else: [\(2.12\)](#) for continuous time, the offspring law or transition matrix for discrete time, and every property is read off from that specification by one of the methods above. And the voice is deliberately abstract: the mathematics of a compartment that may rupture does not depend on what the compartment contains, and the chapter is written so that later chapters are free to supply the biology.

2.2 Markov chains

The objects of this thesis are Markov chains, in discrete and continuous time, homogeneous and driven, standing alone and coupled to deterministic dynamics. This section defines the process classes that recur later: discrete-time chains, through random walks and Galton–Watson processes; continuous-time chains, through the Poisson, birth–death and birth–death–catastrophe processes; time-inhomogeneous chains, through the logistic speciation model; and coupled ODE–CTMC systems, through the rupture-into-a-medium schematic. The methods for analysing them are the subject of [sections 2.3](#) and [2.3.3](#).

2.2.1 Discrete-time Markov chains

A *discrete-time Markov chain* is a sequence of random variables $\{X_n : n = 0, 1, 2, \dots\}$ taking values in a countable state space $\mathcal{S}$ whose future depends on its past only through its present: for all n and all states,

$$\mathbb{P}(X_{n+1} = j \mid X_0, \dots, X_n) = \mathbb{P}(X_{n+1} = j \mid X_n). \quad (2.1)$$

The chain is *time homogeneous* when the right-hand side depends on i, j but not on n , in which case the one-step transition probabilities

$$p_{ij} = \mathbb{P}(X_{n+1} = j \mid X_n = i)$$

assemble into the transition matrix $P = (p_{ij})$, whose rows sum to one. The Chapman–Kolmogorov identity

$$\mathbb{P}(X_{n+m} = j \mid X_n = i) = \sum_{k \in \mathcal{S}} \mathbb{P}(X_n = k \mid X_n = i) p_{kj}^{(m)}$$

shows that the n -step transition probabilities are the entries of the matrix power P^n . Every model of this thesis with discrete generations is a chain of this kind, and the two examples that follow — random walks and branching processes — are the two species of it that recur throughout.

2.2.1.1 Simple random walks

Let $\xi_1, \xi_2, \dots$ be independent random variables with $\mathbb{P}(\xi_k = 1) = p$ and $\mathbb{P}(\xi_k = -1) = q = 1 - p$, and set

$$X_n = X_0 + \xi_1 + \dots + \xi_n. \quad (2.2)$$

The sequence $\{X_n\}$ is a Markov chain on $\mathbb{Z}$ with transitions $p_{i,i+1} = p$, $p_{i,i-1} = q$: from any state the walk moves one step right with probability p and one step left with probability q . Its n -step law is explicit. If R of the n increments are $+1$, then $X_n = X_0 + R - (n - R) = X_0 + 2R - n$, and $R \sim \text{Binomial}(n, p)$; hence

$$\mathbb{P}(X_n = X_0 + 2r - n) = \binom{n}{r} p^r q^{n-r}, \quad r = 0, 1, \dots, n. \quad (2.3)$$

The first two moments follow at once:

$$\mathbb{E}(X_n) = X_0 + (p - q)n, \quad \text{Var}(X_n) = 4pqn.$$

The walk therefore drifts linearly when $p \neq q$ and diffuses, with variance growing like n , when $p = q = \frac{1}{2}$. Figure 2.1 shows the exact law at $n = 40$ for three drifts.

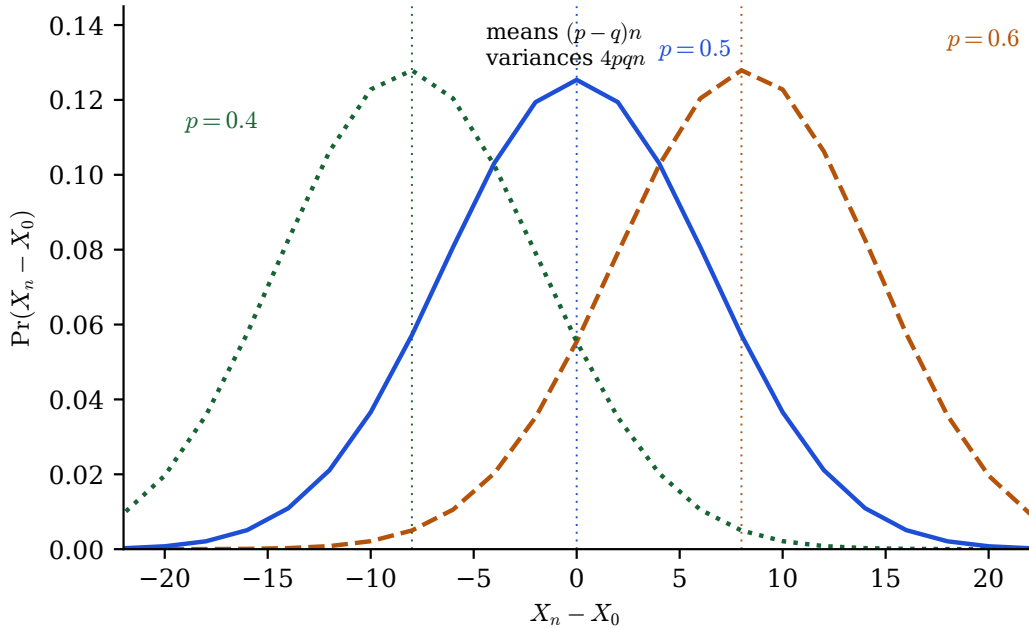


Figure 2.1: Exact position law (2.3) of the simple random walk at $n = 40$, for $p = 0.4$, 0.5 and 0.6 . The distribution is binomial on the reachable lattice, centred on the drift $(p - q)n$ with variance $4pqn$; dotted vertical lines mark the three means. The symmetric walk is the borderline case between the two drift regimes, and its recurrence is the discrete prototype of the critical behaviour of section 2.C.

Two facts about the symmetric walk are used as standing intuition, and both are proved by the methods of [sections 2.3.1](#) and [2.3.2](#) rather than by [\(2.3\)](#) itself. First, the symmetric walk on $\mathbb{Z}$ is *recurrent*: it returns to its starting point, and indeed visits every state, with probability one, although the expected return time is infinite. Second, and this is the prototype of every hitting-probability calculation in this thesis, the probability that the walk reaches a level b before a level $a < X_0 < b$ is determined by a first-step recursion, solved in [section 2.3.1](#). The walk is thus the simplest arena in which the two questions of [section 2.1](#) — where the process goes, and whether it gets there at all — come apart.

2.2.1.2 Galton–Watson processes

The Galton–Watson process is a discrete-time Markov chain on the non-negative integers in which every member of a population independently either leaves offspring or does not, at each generational increment. It is the oldest model of stochastic branching [\[1, 2\]](#), and it is the process on which most of the rest of this thesis is built. Only the definition and the one-step structure are set out here; the critical theory is developed in [section 2.C](#), and the analysis of its quasi-stationary amplitude is the subject of Chapter 3 (*The constant $A(p)$*).

Suppose each individual produces, independently of every other, a number of offspring distributed as the random variable L , with $\mathbb{P}(L = k) = p_k$ and mean $\mu = \mathbb{E}(L)$, and let Z_n be the size of the n th generation, started from a single progenitor, $Z_0 = 1$. Then

$$Z_{n+1} = \sum_{i=1}^{Z_n} L_i, \quad (2.4)$$

the L_i being independent copies of L , which makes $\{Z_n\}$ a Markov chain with transitions from i given by the i -fold convolution of the offspring law. Conditioning on the previous generation gives $\mathbb{E}(Z_n) = \mu^n$. Writing $\phi(z) = \mathbb{E}(z^L)$ for the offspring probability generating function, the composition rule of [section 2.3.1](#) shows that the probability generating function of Z_n is the n -fold composition of ϕ with itself, and in particular the extinction probability $u_n = \mathbb{P}(Z_n = 0)$ satisfies the iteration

$$u_{n+1} = \phi(u_n), \quad u_0 = 0, \quad (2.5)$$

which is the object that [section 2.3.1](#) takes up.

Throughout this thesis we specialise to the binary, or *simple*, case, in which an individual either divides into two, with probability p , or dies, with probability $1 - p$. Then $\mu = 2p$, the offspring probability generating function is

$$\phi(z) = pz^2 + (1 - p), \quad (2.6)$$

and [\(2.5\)](#) becomes $u_{n+1} = pu_n^2 + 1 - p$. In terms of the survival probability $S_n = 1 - u_n = \mathbb{P}(Z_n > 0)$ the same iteration reads

$$S_{n+1} = 2pS_n - pS_n^2, \quad S_0 = 1. \quad (2.7)$$

The process is *subcritical* when $2p < 1$, *critical* when $p = \frac{1}{2}$ and *supercritical* when $2p > 1$; extinction is certain in the first two cases and occurs with probability strictly less than one in the third ([section 2.C](#)). [Figure 2.2](#) shows realisations either side of the critical point.

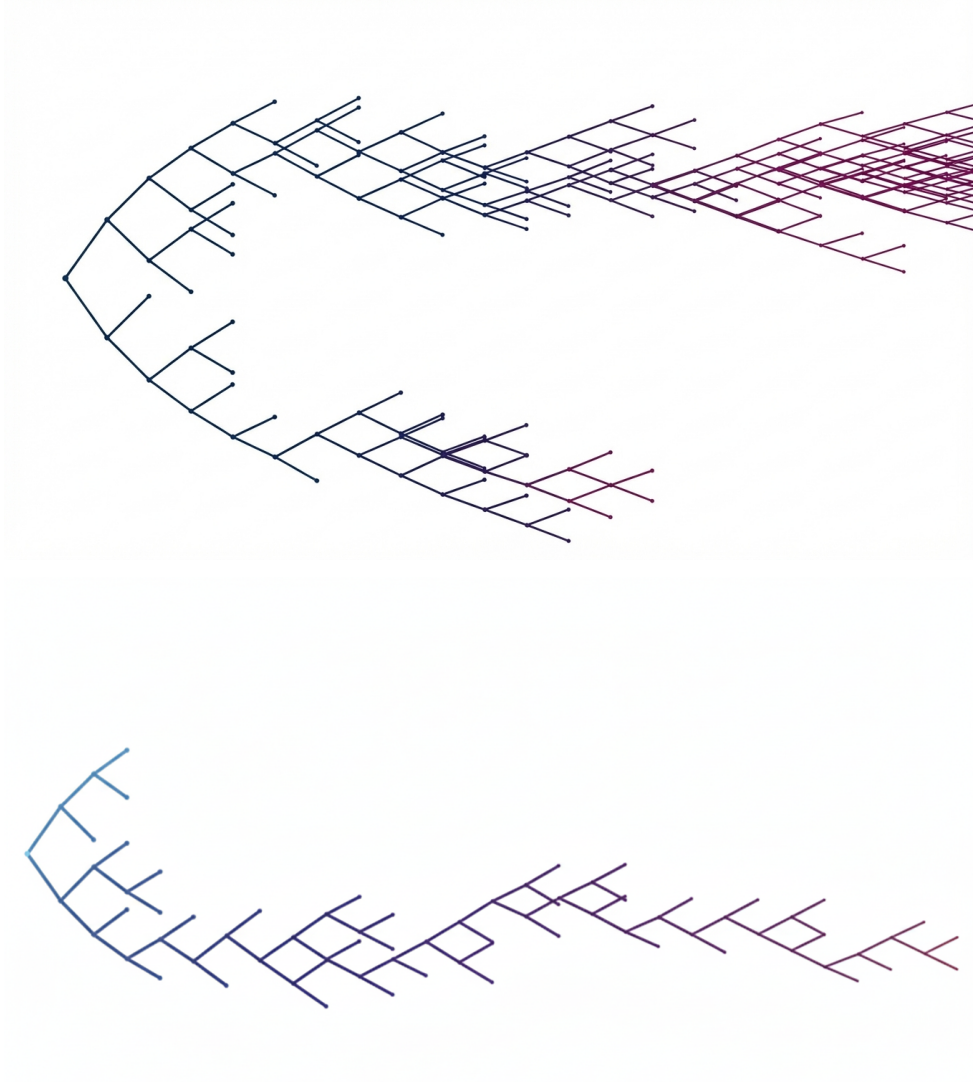


Figure 2.2: Realisations of the simple Galton–Watson process, drawn as genealogies with generation increasing to the right. Top: supercritical, $p = 0.58$, where the number of lines grows without bound. Bottom: subcritical, $p = 0.47$; the same construction, but the tree terminates. Colour encodes generation number.

2.2.2 Continuous-time Markov chains

2.2.2.1 Exponential clocks

Every continuous-time model in this thesis is built from events whose rate of occurrence does not depend on how long the system has been waiting for them. That single assumption forces the exponential distribution: a random variable T is exponentially distributed with rate $\theta > 0$, written $T \sim \text{Exp}(\theta)$, if it has density $\theta e^{-\theta t}$ on $t \geq 0$, mean $1/\theta$ and variance $1/\theta^2$. The property that matters is memorylessness: for $t_1, t_2 > 0$,

$$\mathbb{P}(T > t_1 + t_2 \mid T > t_1) = \frac{e^{-\theta(t_1+t_2)}}{e^{-\theta t_1}} = e^{-\theta t_2} = \mathbb{P}(T > t_2), \quad (2.8)$$

so a clock that has already run for time t_1 is statistically indistinguishable from a fresh one. The exponential law is the only continuous distribution with this property, which is why a process assembled from constant-rate events depends on its present state alone and not on its history.

When several events compete, the following result organises everything that follows. Let $T_1, \dots, T_m$ be independent with $T_i \sim \text{Exp}(\theta_i)$, and let $T = \min_i T_i$ be the time of the first event.

Proposition 2.1 (Competing clocks). *Writing $\Theta = \sum_i \theta_i$,*

$$T \sim \text{Exp}(\Theta), \quad \mathbb{P}(T_k = T) = \frac{\theta_k}{\Theta}, \quad (2.9)$$

and the identity of the winning clock is independent of T .

Proof. For the first claim, $\mathbb{P}(T > t) = \prod_i \mathbb{P}(T_i > t) = e^{-\Theta t}$. For the second, condition on the value of T_k :

$$\mathbb{P}(T_k = T) = \int_0^\infty \theta_k e^{-\theta_k u} \prod_{i \neq k} e^{-\theta_i u} du = \theta_k \int_0^\infty e^{-\Theta u} du = \frac{\theta_k}{\Theta},$$

and truncating the integral at t gives $\mathbb{P}(T_k = T, T > t) = (\theta_k/\Theta)e^{-\Theta t} = \mathbb{P}(T_k = T) \mathbb{P}(T > t)$, which is the asserted independence. $\square$

Proposition 2.1 converts rates into probabilities as soon as one asks *which* of several things happened rather than when, and it is the whole content of the direct simulation method: draw the next event time from $\text{Exp}(\Theta)$, draw the event k with probability θ_k/Θ , advance and repeat [3]. Because the two draws are independent and the exponential law is memoryless, the resulting trajectory has exactly the law of the process. The scheme returns in [section 2.2.4](#), where the rates themselves move between jumps.

2.2.2.2 Generators and forward equations

Let $\{X_t : t \geq 0\}$ take values in a countable state space $\mathcal{S}$, which throughout will be a set of population counts or of tuples of counts. The process is a *continuous-time Markov chain* if, for all $s, t \geq 0$,

$$\mathbb{P}(X_{t+s} = j \mid X_t = i, \{X_u\}_{u < t}) = \mathbb{P}(X_{t+s} = j \mid X_t = i), \quad (2.10)$$

and *time homogeneous* when the right-hand side depends on s but not on t ; time-inhomogeneous chains are the subject of [section 2.2.3](#). A homogeneous chain is specified by its *generator* $Q = (q_{ij})$, whose off-diagonal entries are transition rates,

$$q_{ij} = \lim_{\Delta t \rightarrow 0} \frac{p_{ij}(\Delta t)}{\Delta t} \quad (j \neq i), \quad q_{ii} = -\sum_{j \neq i} q_{ij} =: -q_i, \quad (2.11)$$

so that every row sums to zero. Equivalently,

$$\mathbb{P}(X_{t+\Delta t} = j \mid X_t = i) = q_{ij} \Delta t + o(\Delta t) \quad (j \neq i), \quad (2.12)$$

with complementary probability $1 - q_i \Delta t + o(\Delta t)$ of no transition. By [proposition 2.1](#) the chain leaves state i after a holding time distributed as $\text{Exp}(q_i)$ and then jumps to j with probability q_{ij}/q_i . A state with $q_i = 0$ is *absorbing*: the chain that reaches it stays there forever. Extinction and rupture are both modelled as absorbing states, and the conditional methods of [section 2.3.2](#) concern the behaviour of a chain conditioned on not yet having been absorbed.

Differentiating the Chapman–Kolmogorov identity gives the *forward Kolmogorov equations*,

$$\frac{dp_{ij}(t)}{dt} = \sum_k p_{ik}(t) q_{kj}, \quad \text{that is} \quad \frac{dP}{dt} = P Q, \quad (2.13)$$

which for a fixed initial state and with $p_j(t) = \mathbb{P}(X_t = j)$ become the master equation

$$\frac{dp_j(t)}{dt} = \sum_{k \neq j} q_{kj} p_k(t) - q_j p_j(t) : \quad (2.14)$$

probability flows into state j from every state that can reach it and out of j at total rate q_j . Every model in this thesis is specified by writing down (2.12) and reading off (2.14); the companion *backward* equations and their uses are developed in [section 2.3.2.1](#).

2.2.2.3 The Poisson process

The Poisson process is the CTMC from which all others are built, and the prototype of a pure-birth process. Its state space is $\{0, 1, 2, \dots\}$ and its only transitions are $n \rightarrow n+1$ at constant rate λ , so $q_{n,n+1} = \lambda$ and $q_n = \lambda$ for every n : the holding time in every state is $\text{Exp}(\lambda)$, independent of everything else by (2.8), and the process counts the arrivals of a stream of events occurring one at a time at constant rate.

The master equation (2.14) reads

$$\frac{dp_n(t)}{dt} = \lambda p_{n-1}(t) - \lambda p_n(t), \quad p_n(0) = \mathbf{1}_{\{n=0\}},$$

which solves inductively to the Poisson law

$$N_t \sim \text{Poisson}(\lambda t), \quad \mathbb{P}(N_t = n) = e^{-\lambda t} \frac{(\lambda t)^n}{n!}, \quad (2.15)$$

with $\mathbb{E}(N_t) = \text{Var}(N_t) = \lambda t$. Two immediate readings of (2.15) are worth keeping. The number of events in disjoint time intervals are independent random variables, because the exponential holding times restart afresh after every event; and the jump times of any time-homogeneous CTMC whose exit rate happens to stay constant — for instance a pure-birth process before its first death — form a Poisson process.

The Poisson process is also the limit object behind thinning and superposition: retaining each event of a rate- λ process independently with probability p leaves a Poisson process of rate $p\lambda$, and independent Poisson streams of rates λ_1, λ_2 combine to one of rate $\lambda_1 + \lambda_2$ — both direct consequences of (2.15) and independence. These facts are used quietly below whenever a stream of events is split by type or merged.

2.2.2.4 Birth–death processes

The linear birth–death process is the continuous-time counterpart of the Galton–Watson process and the running example of [section 2.3.2](#). Let X_t count individuals, each of which independently gives birth at rate λ and dies at rate μ . The non-zero rates out of state n are

$$q_{n,n+1} = \lambda n, \quad q_{n,n-1} = \mu n, \quad (2.16)$$

so that $q_n = (\lambda + \mu)n$ and $n = 0$ is absorbing: the rates are linear in the count, which is what makes the process exactly tractable by the methods of [section 2.3.2](#). The master equation reads

$$\frac{dp_n(t)}{dt} = \lambda(n-1)p_{n-1}(t) + \mu(n+1)p_{n+1}(t) - (\lambda + \mu)n p_n(t). \quad (2.17)$$

With the probability generating function $G(z, t) = \mathbb{E}(z^{X_t})$, multiplying (2.17) by z^n , summing over n and shifting indices collapses the infinite family of state probabilities to a single partial differential equation,

$$\frac{\partial G}{\partial t} = (\lambda z^2 - (\lambda + \mu)z + \mu) \frac{\partial G}{\partial z} = (\lambda z - \mu)(z - 1) \frac{\partial G}{\partial z}, \quad G(z, 0) = z^N, \quad (2.18)$$

for a process started from N individuals. This conversion — master equation to first-order generating-function equation — is the template for every generating-function calculation in this thesis, and its solution by the method of characteristics is prepared in [section 2.3.3](#). The result, obtained there, is Kendall's formula [4]:

$$G(z, t) = \left(\frac{\mu(1-z) - (\mu - \lambda z)e^{-(\lambda - \mu)t}}{\lambda(1-z) - (\mu - \lambda z)e^{-(\lambda - \mu)t}} \right)^N, \quad \lambda \neq \mu, \quad (2.19)$$

from which the mean $\mathbb{E}(X_t) = Ne^{(\lambda - \mu)t}$ and the extinction probability by time t ,

$$p_0(t) = \begin{cases} \left(\frac{\mu - \mu e^{(\mu - \lambda)t}}{\lambda - \mu e^{(\mu - \lambda)t}} \right)^N, & \lambda \neq \mu, \\ \left(\frac{\lambda t}{1 + \lambda t} \right)^N, & \lambda = \mu, \end{cases} \quad (2.20)$$

follow by differentiation and by setting $z = 0$. Letting $t \rightarrow \infty$ recovers the familiar threshold: extinction is certain when $\lambda \leq \mu$ and occurs with probability $(\mu/\lambda)^N$ when $\lambda > \mu$. The quadratic coefficient in (2.18) factorises with roots $z = 1$ and $z = \mu/\lambda$; the same factorisation, in a different variable, reappears in [section 2.G](#).

A useful extension, which later chapters use, lets the rates vary slowly with the population itself. If λ, μ are replaced by $\lambda(n), \mu(n)$ — for instance $\lambda(n) = \lambda_0(1 - n/K)$ to model crowding — the chain remains Markov but is no longer exactly solvable by the closed route above; mean-field treatments replace n by $\mathbb{E}(X_t)$ and recover deterministic density-dependent equations [5]. The time-inhomogeneous linear case, in which the rates are prescribed functions of t alone, remains tractable and is taken up next.

2.2.2.5 Birth–death–catastrophe processes

Alongside extinction by death, several later chapters need a second absorbing mechanism: an abrupt event that removes the entire population at once. The continuous-time birth–death–catastrophe process adds that channel to [section 2.2.2.4](#).

Definition 2.2 (Birth–death–catastrophe process). Let X_t count individuals in a compartment. Each individual independently gives birth at rate λ and dies at rate μ , and the compartment additionally undergoes a catastrophe — a rupture that removes the entire population at once —

at a rate ρ per individual, so that the total catastrophe rate in state n is ρn . Writing $\dagger$ for the post-catastrophe state, the non-zero transition rates out of $n \geq 1$ are

$$q_{n,n+1} = \lambda n, \quad q_{n,n-1} = \mu n, \quad q_{n,\dagger} = \rho n, \quad (2.21)$$

and both $n = 0$ and $\dagger$ are absorbing.

Two features distinguish [definition 2.2](#) from [section 2.2.2.4](#), and both matter later. The process has *two* absorbing states, reached by mechanisms that are distinguishable and not interchangeable: extinction empties the compartment gradually, catastrophe empties it in one event and releases its contents. And the catastrophe rate is proportional to the population, so the compartment becomes more likely to rupture the fuller it is.

Restricting attention to realisations that have not yet ruptured and writing $G(z, t) = \mathbb{E}\left(z^{X_t} \mathbf{1}\{X_t \neq \dagger\}\right)$, the master equation for [\(2.21\)](#) yields

$$\frac{\partial G}{\partial t} = (\lambda z^2 - (\lambda + \mu + \rho)z + \mu) \frac{\partial G}{\partial z}, \quad G(z, 0) = z^N, \quad (2.22)$$

which differs from [\(2.18\)](#) only in the coefficient of z . The quadratic no longer factorises with a root at $z = 1$, and that single change is responsible for most of what is interesting about the process: probability is not conserved on the non-ruptured states, and $G(1, t)$ is the survival probability rather than one. The discrete-generation counterpart, its exact no-rupture identities, and the killed-generator formulation of quasi-stationarity for this process are developed in [section 2.F](#); rates, moments and state probabilities for [definition 2.2](#) itself are treated in the later chapters on birth–death–catastrophe processes, with the two-type extension in the chapter on multi-type processes and the release of a ruptured compartment’s contents into a shared medium in the chapter on compartment rupture.

2.2.3 Time-inhomogeneous processes

In every chain defined so far the rates are fixed numbers. Several later chapters need rates that vary with the clock: an environment that deteriorates, a resource that depletes, a speciation rate that declines as the available niche fills. This section sets up the time-inhomogeneous framework and works through the standard example, the logistic speciation model, which is exactly soluble at the level required here.

2.2.3.1 Chains with time-dependent rates

A *time-inhomogeneous* continuous-time Markov chain keeps the Markov property [\(2.10\)](#) but allows the transition rates to depend on absolute time: in place of [\(2.12\)](#),

$$\mathbb{P}(X_{t+\Delta t} = j \mid X_t = i) = q_{ij}(t) \Delta t + o(\Delta t) \quad (j \neq i). \quad (2.23)$$

The forward equations acquire a time-dependent generator,

$$\frac{dP}{dt} = P Q(t), \quad \frac{dp_j(t)}{dt} = \sum_{k \neq j} q_{kj}(t) p_k(t) - q_j(t) p_j(t), \quad (2.24)$$

and the holding time in state i entered at time s is no longer exponential with a constant rate but has survival function $\exp(-\int_s^t q_i(u) du)$. Two structural losses should be noted. The transition

probabilities no longer form a semigroup — they depend on the starting time as well as the elapsed time — and there is in general no stationary distribution to converge to, because the target itself is moving. What survives, and this is what the methods of [section 2.3.2](#) exploit, is linearity: (2.24) is still a linear system, generating functions still convert it to a first-order equation, and first-step arguments still close, now with t -dependent coefficients.

2.2.3.2 The time-inhomogeneous linear birth–death process

The linear birth–death process with rates $\lambda(t)$, $\mu(t)$ — each individual gives birth at rate $\lambda(t)$ and dies at rate $\mu(t)$, the rates being prescribed functions of time — is the workhorse example. The master equation is (2.17) with t -dependent coefficients, and the generating function satisfies

$$\frac{\partial G}{\partial t} = (\lambda(t)z^2 - (\lambda(t) + \mu(t))z + \mu(t)) \frac{\partial G}{\partial z}. \quad (2.25)$$

Two quantities remain explicit for arbitrary integrable rates. The mean, by the same conditioning argument as in the homogeneous case, solves $\frac{d\mathbb{E}(X_t)}{dt} = (\lambda(t) - \mu(t))\mathbb{E}(X_t)$, hence

$$\mathbb{E}(X_t) = N \exp\left(\int_0^t (\lambda(s) - \mu(s)) ds\right). \quad (2.26)$$

The extinction probability $u(t) = \mathbb{P}(X_t = 0)$, started from one individual, solves the Riccati equation

$$\frac{du}{dt} = \mu(t) - (\lambda(t) + \mu(t))u + \lambda(t)u^2, \quad u(0) = 0, \quad (2.27)$$

which is the backward equation of [section 2.3.2.1](#) with time-dependent coefficients: in the first small interval either a death occurs, contributing $\mu(t)\Delta t$, or a birth occurs, contributing $\lambda(t)\Delta t u(t)^2$ since both daughter lineages must then die out, or nothing occurs. The full generating function retains the fractional-linear form of (2.19), with constant coefficients replaced by solutions of the same Riccati equation; when (2.27) cannot be solved in closed form, it is at least a scalar equation and is integrated numerically without difficulty.

2.2.3.3 The logistic speciation model

The standard macroevolutionary application of [section 2.2.3.2](#) lets the speciation rate decline with diversity [5]. Let X_t count species, each of which speciates at rate

$$\lambda(n) = \lambda_0 \left(1 - \frac{n}{K}\right) \quad (2.28)$$

when the standing diversity is n , and goes extinct at constant rate μ , with $\lambda_0 > \mu$ and K the niche capacity. The chain with the genuinely state-dependent rate (2.28) is nonlinear and has no closed solution; the standard tractable surrogate replaces the random diversity in (2.28) by the deterministic mean-field trajectory, yielding a *time-inhomogeneous* linear process in the sense of [section 2.2.3.2](#). The substitution is a modelling step, not an identity, and is flagged as such wherever the model is used below.

The mean-field diversity $\bar{N}(t)$ solves

$$\frac{d\bar{N}}{dt} = (\lambda(\bar{N}) - \mu)\bar{N} = r\bar{N} \left(1 - \frac{\bar{N}}{N_*}\right), \quad r = \lambda_0 - \mu, \quad N_* = K \left(1 - \frac{\mu}{\lambda_0}\right), \quad (2.29)$$

the logistic equation, whose carrying point N_* is the diversity at which speciation balances extinction. Starting from $\bar{N}(0) = N_0$,

$$\bar{N}(t) = \frac{N_*}{1 + Be^{-rt}}, \quad B = \frac{N_*}{N_0} - 1, \quad (2.30)$$

and the surrogate speciation rate $\lambda(t) = \lambda_0(1 - \bar{N}(t)/K)$ is

$$\lambda(t) = \mu + \frac{rBe^{-rt}}{1 + Be^{-rt}}, \quad \lambda(0) = \lambda_0 \left(1 - \frac{N_0}{K}\right), \quad (2.31)$$

relaxing monotonically from $\lambda(0)$ down to μ (figure 2.3). The accumulated net growth needed in (2.26) evaluates in closed form,

$$\int_0^t (\lambda(s) - \mu) ds = \ln \left(\frac{1 + B}{1 + Be^{-rt}} \right), \quad (2.32)$$

so the mean diversity of the surrogate process is

$$\mathbb{E}(X_t) = N_0 \frac{1 + B}{1 + Be^{-rt}} \xrightarrow[t \rightarrow \infty]{} N_0(1 + B) = N_*, \quad (2.33)$$

which recovers the deterministic equilibrium, as a linear mean-field surrogate should. Its extinction probability is obtained by integrating (2.27) with the rate (2.31). Two limiting behaviours are worth noting: at early times $\lambda(t) \approx \lambda(0)$ and the process is indistinguishable from a homogeneous supercritical one, while at late times $\lambda(t) \rightarrow \mu$ from above and the chain approaches criticality — the regime whose discrete-generation difficulties are documented in section 2.C.

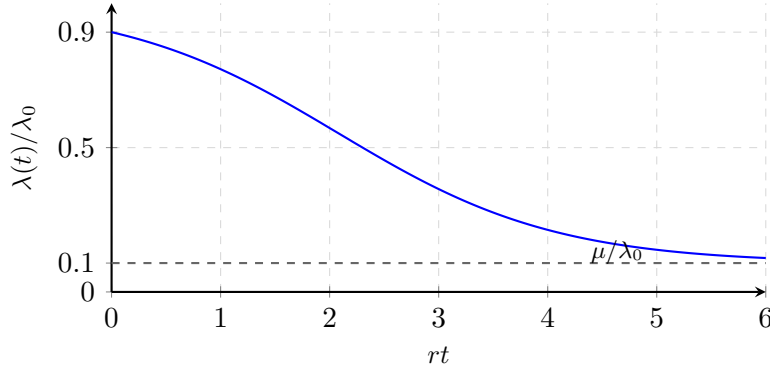


Figure 2.3: The diversity-dependent speciation rate (2.31) of the logistic speciation model, in units of λ_0 and against the rescaled time rt , for $\mu/\lambda_0 = 0.1$ and $N_*/N_0 = 9$ (so $B = 8$ and $\lambda(0) = 0.9\lambda_0$). The rate relaxes from $\lambda(0)$ down to the extinction rate μ as the mean-field diversity (2.30) fills the equilibrium N_* ; the integrated intensity (2.32) controls the mean (2.33).

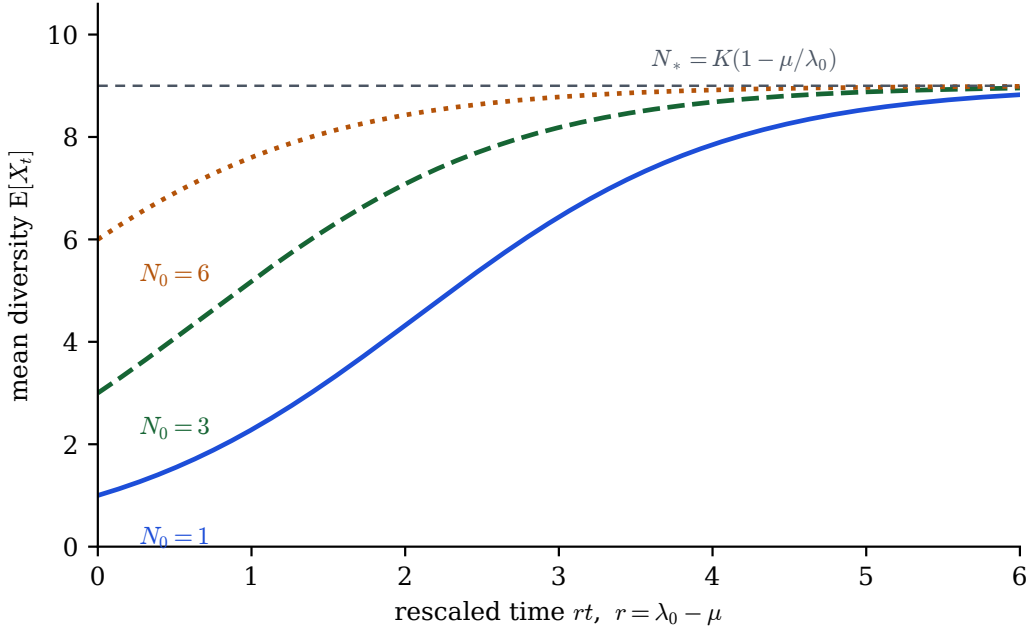


Figure 2.4: Mean diversity (2.33) of the logistic speciation surrogate for $N_0 = 1, 3, 6$, with the chapter’s parameters $\mu/\lambda_0 = 0.1$, $K = 10$ (so $N_* = 9$), plotted against the rescaled time rt . Every founding diversity relaxes monotonically to the equilibrium $N_* = K(1 - \mu/\lambda_0)$, and the mean trajectory coincides with the deterministic mean-field solution (2.30) — the surrogate is linear, so its mean closes exactly. Small founding cohorts spend a long early interval in near-exponential growth before the diversity dependence binds.

2.2.4 Coupled ODE–CTMC systems

The chains of sections 2.2.2 and 2.2.3 have rates that are either fixed numbers or prescribed functions of time. Several later chapters need a third possibility: rates that are functions of quantities evolving according to their own deterministic dynamics, where those dynamics in turn depend on the state of the chain. This section fixes the framework for such systems. It is conceptual — the framework and one schematic example — and the concrete models built on it are developed in the later chapters on compartment rupture and shared media.

2.2.4.1 The framework

Let the state of the system be a pair (Y_t, X_t) , where X_t is a CTMC on a countable state space $\mathcal{S}$, interpreted as the discrete configuration of a population or of a collection of compartments, and $Y_t \in \mathbb{R}^d$ is a vector of continuous variables — concentrations, accumulated damage, the contents of a shared medium. The coupling has two directions.

Definition 2.3 (Coupled ODE–CTMC system). Between jumps of X , the continuous state evolves by

$$\frac{dY}{dt} = f(Y_t, X_t), \quad (2.34)$$

with $f : \mathbb{R}^d \times \mathcal{S} \rightarrow \mathbb{R}^d$. When $X_t = i$, the chain jumps to j at rate $q_{ij}(Y_t)$, so that the generator

depends on the current continuous state:

$$\mathbb{P}(X_{t+\Delta t} = j \mid X_t = i, Y_t) = q_{ij}(Y_t) \Delta t + o(\Delta t). \quad (2.35)$$

The dependence may operate in either direction alone, or in both at once.

The joint process (Y_t, X_t) is Markov whenever the ODE flow is well posed — say $f(\cdot, i)$ locally Lipschitz for each i — and the rates are measurable and bounded on bounded sets. It belongs to the class of *piecewise-deterministic Markov processes*: the randomness enters only through the jump mechanism, and between jumps the trajectory is a solution of an ordinary differential equation [6]. Two limiting cases connect the definition to earlier material. If q_{ij} does not actually depend on Y , [definition 2.3](#) reduces to a CTMC forcing an ODE; if f does not depend on X , it reduces to an ODE whose coefficients are modulated by an external chain. The genuinely coupled case, which is the one that appears in this thesis, uses both dependences at once ([figure 2.5](#)).

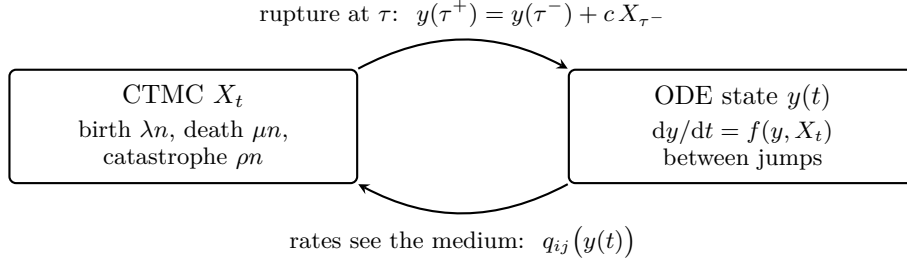


Figure 2.5: The two directions of the coupling in [definition 2.3](#). The jump process X_t drives the flow through its state (and, at a rupture, through an impulsive release into the medium), while the continuous state $y(t)$ modulates the transition rates of the chain. Between jumps the trajectory is a solution of an ODE; randomness enters only at the jumps, which is the piecewise-deterministic structure of [6].

The one-way versions are familiar in the applications. A chain forcing an ODE is what one obtains when discrete events feed a continuous reservoir: each jump of X changes the source term of (2.34), and the reservoir integrates the resulting forcing between jumps. An ODE modulating a chain is the natural description of an environment: a drug concentration that decays deterministically while the per-capita death rate of a population is a function of it.

2.2.4.2 Simulation and analysis

The construction of [proposition 2.1](#) extends to [definition 2.3](#) with one qualification: the rates are no longer constant between events, because Y_t moves. Given the current state (Y, i) , integrate (2.34) with $X \equiv i$ to obtain Y_s for $s \geq t$; the total jump rate $q_i(Y_s) = \sum_j q_{ij}(Y_s)$ is then a known function of s , and the next jump time T has survival function

$$\mathbb{P}(T > s) = \exp \left(- \int_t^s q_i(Y_u) du \right),$$

exactly as in [section 2.2.3.1](#); the jump destination is drawn with probability $q_{ij}(Y_T)/q_i(Y_T)$, the state is updated, and the procedure repeats. When q_i is bounded, the integral can be inverted by rejection on a constant majorant; otherwise it is quadrature, and the scheme is the standard event-driven simulator for piecewise-deterministic processes.

Analytically, two routes are available. If the coupling is weak, one replaces Y_t by its mean-field trajectory, obtaining a time-inhomogeneous chain in the sense of [section 2.2.3](#) — the logistic speciation model of [section 2.2.3.3](#) is precisely that closure — or replaces X_t by its mean, obtaining a closed ODE. When the coupling is strong neither closure is exact, and the working objects are the joint expectations, which satisfy hybrid moment equations mixing the generator of the chain with the drift f ; those are developed as needed in the chapters on rupture and shared media.

2.2.4.3 A schematic example: rupture feeding a shared medium

The example that organises the later chapters is a compartment that can rupture into a shared medium. Let X_t be the contents of the compartment, evolving as the birth–death–catastrophe process of [definition 2.2](#), and let $y(t) \geq 0$ be the concentration of material in the shared medium. Between ruptures the medium loses material at rate δ ,

$$\frac{dy}{dt} = -\delta y,$$

and at the rupture time τ the entire contents of the compartment are released into the medium in one jump,

$$y(\tau+) = y(\tau-) + c X_{\tau-}.$$

The reverse direction of the coupling is equally present: the absorption rate at which surviving compartments take up material from the medium, and the catastrophe rate ρ itself, may both be functions of the current concentration $y(t)$, so that a fuller medium accelerates uptake and can destabilise what remains. Written in the notation of [definition 2.3](#), this is a system in which f depends on X (release occurs only through the catastrophe channel) and q_{ij} depends on Y (the rates see the medium), so both arrows of the definition are active.

Two features of this schematic are worth isolating, because they recur in every instantiation of it. The ODE variable y is *impulsive*: it evolves smoothly between jumps and discontinuously at them, so its natural description is a flow plus jump maps rather than a single differential equation. And the quantity released at rupture, $X_{\tau-}$, is the state of the chain at a stopping time, not a parameter: the rupture cohort is whatever the catastrophe finds, and its distribution is the quasi-stationary-type problem of [section 2.3.2](#) evaluated at a random time. Both features are why the models built on [definition 2.3](#) are analysed with generating-function and first-step machinery rather than closed by simulation alone.

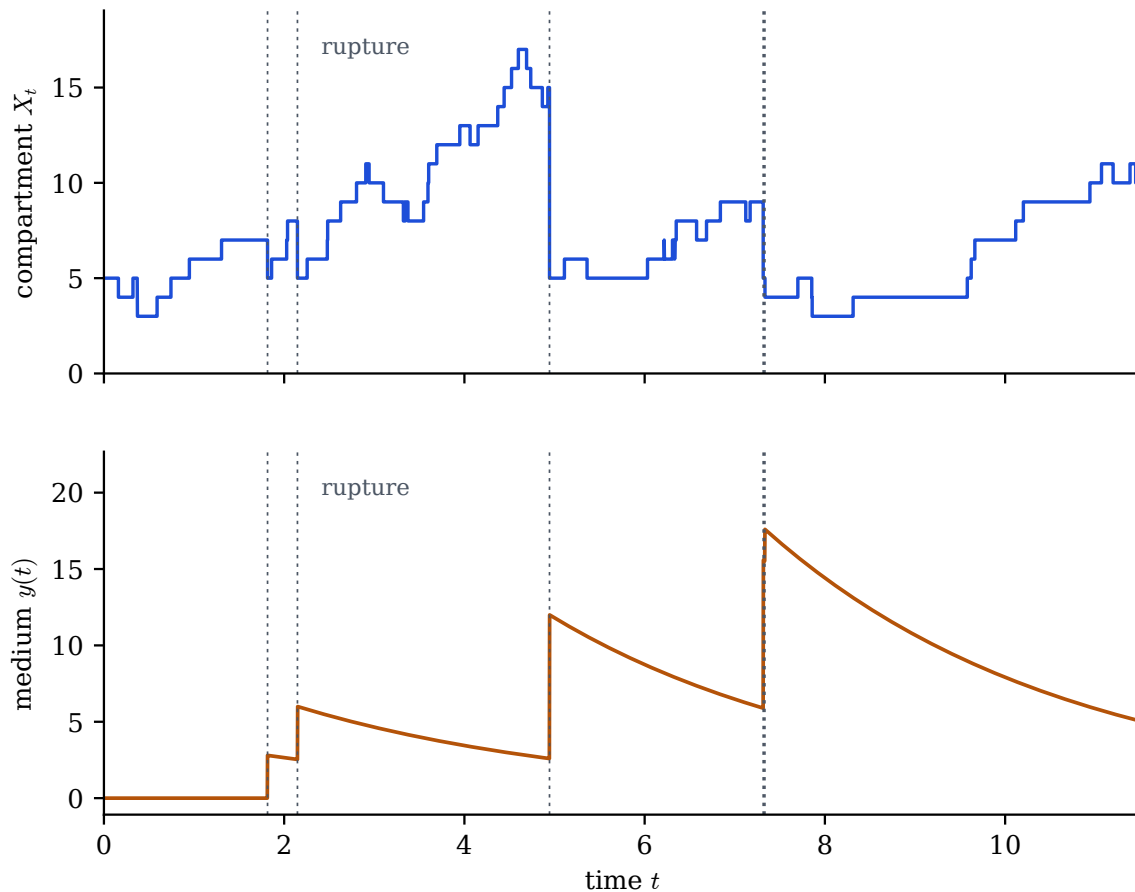


Figure 2.6: A single exact event-driven realisation of the coupled rupture system of [section 2.2.4.3](#), with $\lambda = 0.55$, $\mu = 0.25$, $\rho = 0.08$, $N_0 = 5$, $\delta = 0.30$ and $c = 0.4$; for illustration the compartment is re-seeded with N_0 individuals after each rupture. Top: the compartment contents X_t , growing between ruptures and collapsing at them. Bottom: the shared medium $y(t)$, decaying exponentially between ruptures and receiving the impulsive release $cX_{\tau-}$ at each one; the jumps grow with the cohort the catastrophe finds. Dotted vertical lines mark the rupture times, aligned across the panels.

2.3 Methods for Markov chains

The process classes of the preceding section are analysed with a small stock of methods that recur throughout the thesis. This section introduces the stock in two halves — discrete-time methods briefly, continuous-time methods at length — and [section 2.3.3](#) adds the partial-differential-equation method that the generating-function equations require. The division is by time domain rather than by problem: every method below has a counterpart on the other side of the divide, and part of the point of the chapter is to make the correspondences visible.

2.3.1 Discrete-time methods

Three instruments do almost all of the discrete-time work in this thesis: probability generating functions, functional iteration, and first-step analysis. This section fixes them; the deeper discrete-time theory needed later — the analysis of the quasi-stationary amplitude of the binary Galton–Watson process by Koenigs-function methods — is the subject of Chapter 3 (*The constant $A(p)$*)

and is not anticipated here.

2.3.1.1 Generating functions and composition

For a random variable X taking values in $\{0, 1, 2, \dots\}$ with $p_k = \mathbb{P}(X = k)$, the probability generating function is

$$\phi(z) = \mathbb{E}(z^X) = \sum_{k=0}^{\infty} p_k z^k, \quad (2.36)$$

which converges at least for $|z| \leq 1$. The series determines the distribution — $p_k = \phi^{(k)}(0)/k!$ — and moments are read off at $z = 1$:

$$\phi(1) = 1, \quad \phi'(1) = \mathbb{E}(X), \quad \phi''(1) = \mathbb{E}(X(X-1)). \quad (2.37)$$

Two structural properties do most of the work. If X and Y are independent then $\phi_{X+Y} = \phi_X \phi_Y$, so sums become products. And if N is an independent count with probability generating function ψ , the random sum $X_1 + \dots + X_N$ of independent copies of X has probability generating function $\psi \circ \phi$: composition of generating functions is what makes branching processes tractable, and it is the reason the Galton–Watson recursion closes on a single function. For several counts the definition extends to the multivariate probability generating function $\mathbb{E}(z_1^{X_1} \dots z_d^{X_d})$, with each marginal recovered by setting the remaining variables to one; multi-type generating functions of this kind are the natural language of the two-type extensions treated in the chapter on multi-type processes.

2.3.1.2 Functional iteration

Applied to the Galton–Watson process, composition says that the probability generating function of Z_n is the n -fold composition of the offspring probability generating function ϕ with itself, and that the extinction probability $u_n = \mathbb{P}(Z_n = 0)$ satisfies

$$u_{n+1} = \phi(u_n), \quad u_0 = 0.$$

The limit $u_\infty = \lim_n u_n$ — extinction is an absorbing event, so the limit exists — is a fixed point of ϕ , and the standard argument shows it is the *smallest* fixed point in $[0, 1]$ [1]: the sequence u_n increases, so any fixed point ζ with $u_0 \leq \zeta$ bounds it from above, and the smallest such bound is itself a fixed point by continuity. For the binary law $\phi(z) = pz^2 + 1 - p$ the fixed points are 1 and $(2p - 1)/p$; the threshold $p = \frac{1}{2}$ falls out at once, and the fine structure at and below criticality — the power-law survival, the infinite expected lifetime — is developed in [section 2.C](#).

The same iteration governs survival in the presence of killing. A discrete birth–death process subject to per-step catastrophe, analysed in [section 2.F](#), carries a weighted generating function satisfying the closed recursion $F_{n+1} = c\phi(F_n)$, and the joint survival-without-rupture probability is recovered from its evaluations at 1 and 0. Iteration with a prefactor is no harder than iteration without one, and the distinction between the two recursions is exactly the distinction between survival and non-catastrophe.

2.3.1.3 First-step analysis

When the question is not the full time course but a hitting probability or a mean time, one conditions on the first exit from the current state and solves the resulting self-consistent equation. The gesture is the same each time: one step of randomness, then the Markov property restarts the problem.

The prototype is the gambler's ruin for the simple random walk of [section 2.2.1.1](#). Let h_i be the probability that the walk, started at i with $a < i < b$, reaches b before a . Conditioning on the first step,

$$h_i = p h_{i+1} + q h_{i-1}, \quad h_a = 0, \quad h_b = 1, \quad (2.38)$$

a second-order difference equation. Writing $\Delta_i = h_i - h_{i-1}$, (2.38) says $p \Delta_{i+1} = q \Delta_i$, so the increments are geometric: $\Delta_i = \Delta_1 (q/p)^{i-1}$. Summing and imposing the boundary conditions gives, for $p \neq q$,

$$h_i = \frac{(q/p)^{i-a} - 1}{(q/p)^{b-a} - 1}, \quad \text{and} \quad h_i = \frac{i-a}{b-a} \quad \text{when } p = q. \quad (2.39)$$

Two readings of (2.39) are worth keeping. Letting $b \rightarrow \infty$ with $p < q$ gives the probability $(q/p)^{i-a}$ of ever reaching b from i when the walk drifts downward; the walk with drift away from its target still reaches it with positive probability, but the chance decays geometrically with distance. And the symmetric case is the limit of the asymmetric one: the formula with $p = q$ is what (2.39) becomes as $q/p \rightarrow 1$, not a separate calculation. The continuous-time versions of both readings are derived in [sections 2.3.2.4](#) and [2.3.2.5](#) by the identical gesture, with a rate-weighted first step in place of (2.38). [Figure 2.7](#) shows the three regimes of (2.39).

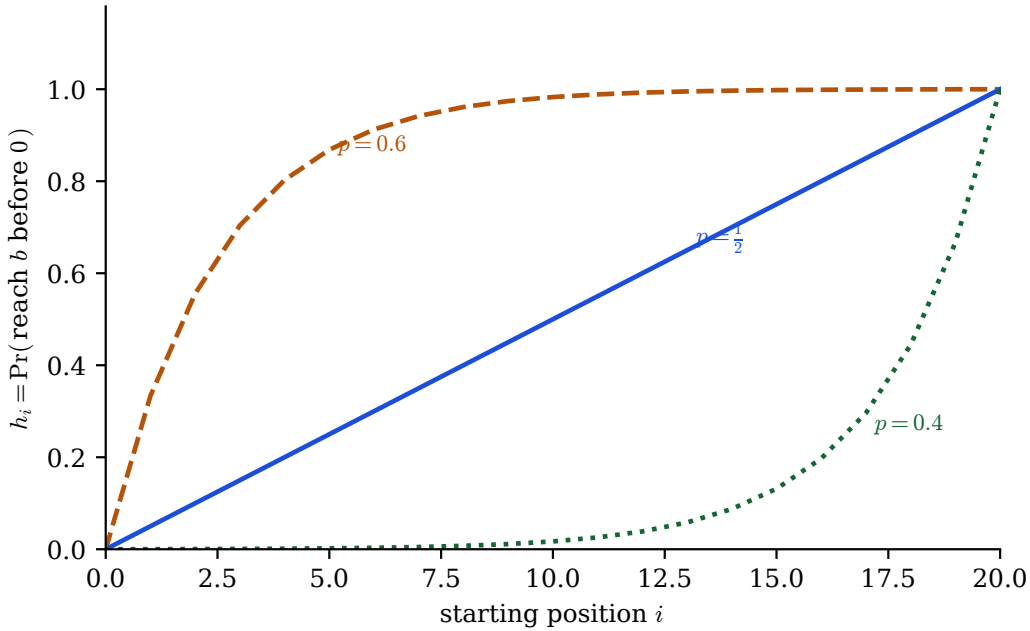


Figure 2.7: Hitting probabilities (2.39) of the gambler's ruin problem with $b = 20$: the probability h_i of reaching b before 0 from starting position i , for $p = 0.4$, $\frac{1}{2}$ and 0.6 . Drift towards the target makes the probability rise steeply and saturate near one; drift away from it confines the mass to the far end of the interval; the symmetric case interpolates linearly. The same first-step difference equation, with q/p replaced by μ/λ , is the continuous-time hitting probability of (2.46).

2.3.2 Continuous-time methods

The continuous-time stock is larger, and this subsection develops it with the linear birth–death process of [section 2.2.2.4](#) as the running example: backward equations, mean population size,

variance and higher moments, hitting probabilities, extinction probabilities, and conditional means. Every item has a discrete-time counterpart in [section 2.3.1](#), and the correspondences are pointed out as they arise. The one item that has no discrete-time analogue at all — the method of characteristics for generating-function equations — is large enough to warrant its own subsection, [section 2.3.3](#).

2.3.2.1 Backward equations

The forward equations of (2.13) condition on the starting state and differentiate in the arrival time. The *backward* equations do the opposite: they condition on what happens in the first small interval after the start, and differentiate in the starting point. Formally,

$$\frac{dP}{dt} = Q P, \quad \frac{dp_{ij}(t)}{dt} = \sum_k q_{ik} p_{kj}(t), \quad (2.40)$$

but their practical value is that for absorption probabilities and mean hitting times they collapse at once to scalar equations. The prototype is the extinction probability of the birth–death process.

Let $u(t)$ be the probability that a birth–death process started from one individual is extinct by time t . In the first interval of length h , either a death occurs, with probability $\mu h + o(h)$, in which case extinction is immediate; or a birth occurs, with probability $\lambda h + o(h)$, in which case both daughter lineages must die out within the remaining time, an event of probability $u(t-h)^2$; or nothing occurs. Letting $h \downarrow 0$,

$$\frac{du}{dt} = \mu - (\lambda + \mu)u + \lambda u^2, \quad u(0) = 0. \quad (2.41)$$

This is a Riccati equation of the same family as the survival recursions of (2.7) and (2.27): its right-hand side factorises as $\lambda(u-1)(u-\mu/\lambda)$, so separation of variables gives

$$\frac{u-1}{u-\mu/\lambda} = \frac{\mu}{\lambda} e^{-(\lambda-\mu)t}$$

after imposing $u(0) = 0$, and solving for u recovers the extinction probability of (2.20) for one individual:

$$u(t) = \frac{\mu - \mu e^{-(\lambda-\mu)t}}{\lambda - \mu e^{-(\lambda-\mu)t}}. \quad (2.42)$$

For N initial individuals the lineages are independent, so the probability is $u(t)^N$. The backward route has produced the whole of (2.20) from a scalar first-order equation, without the generating function; that is the bargain the method offers whenever the question is absorption rather than the full law.

2.3.2.2 Mean population size

The mean of the birth–death process satisfies a closed equation because the rates are linear in the count. Conditioning on the first interval as above, or differentiating the master equation (2.17) after multiplying by n and summing,

$$\frac{d}{dt} \mathbb{E}(X_t) = (\lambda - \mu) \mathbb{E}(X_t), \quad \mathbb{E}(X_t) = N e^{(\lambda-\mu)t}. \quad (2.43)$$

The sign of $\lambda - \mu$ decides the fate of the mean, exactly as the sign of $2p - 1$ decided the fate of the Galton–Watson mean in [section 2.2.1.2](#): exponential growth above criticality, exponential

decay below, and constancy at the critical point $\lambda = \mu$, where, as in the discrete case, constancy of the mean is consistent with certain extinction because rare enormous realisations carry it. The time-inhomogeneous analogue, (2.26), replaces the constant exponent by the accumulated intensity $\int_0^t (\lambda - \mu) ds$.

2.3.2.3 Variance and higher moments

The second moment closes in the same way, because applying the generator to $n(n-1)$ produces at most quadratic terms. Let $f(t) = \mathbb{E}(X_t(X_t - 1))$ for a process started from one individual. One more differentiation of the master equation, or equivalently applying the generator to $n(n-1)$, gives

$$\frac{df}{dt} = 2(\lambda - \mu)f + 2\lambda\mathbb{E}(X_t),$$

which with (2.43) and $f(0) = 0$ solves to

$$f(t) = \frac{2\lambda}{\lambda - \mu} \left(e^{2(\lambda - \mu)t} - e^{(\lambda - \mu)t} \right).$$

Since $\text{Var}(X_t) = f(t) + \mathbb{E}(X_t) - \mathbb{E}(X_t)^2$, the variance of the process started from N individuals is

$$\text{Var}(X_t) = N \frac{\lambda + \mu}{\lambda - \mu} \left(e^{2(\lambda - \mu)t} - e^{(\lambda - \mu)t} \right), \quad (2.44)$$

with the critical limit $\text{Var}(X_t) \rightarrow 2N\lambda t$ as $\mu \rightarrow \lambda$. Two features are worth noting. The variance grows like the *square* of the mean when $\lambda > \mu$, so the coefficient of variation tends to a constant rather than to zero: even a supercritical birth–death process remains genuinely random at late times, unlike its deterministic approximation. And the derivation generalises without new ideas: applying the generator to n^k expresses the k th moment in terms of lower ones, so all moments of the linear process are obtainable from a triangular system.

The quasi-stationary variance — the variance of the Galton–Watson population *conditioned on survival* — is a different and subtler object, because the conditioning is time-dependent; its full derivation is in [section 2.E](#).

2.3.2.4 Hitting probabilities

Hitting probabilities for a CTMC satisfy the continuous-time version of the first-step recursion (2.38). If h_i is the probability of reaching a target set before another, started from i , then conditioning on the first jump — which goes to j at rate q_{ij} and occurs after an $\text{Exp}(q_i)$ wait during which nothing changes — gives

$$\sum_{j \neq i} q_{ij} (h_j - h_i) = 0 \quad (2.45)$$

at every interior state, with h fixed on the targets. For the birth–death process, the probability h_i of reaching level b before level 0, started from $0 < i < b$, therefore satisfies

$$\lambda i (h_{i+1} - h_i) + \mu i (h_{i-1} - h_i) = 0,$$

and the factor of i cancels: the same difference equation as the gambler’s ruin (2.38), with q/p replaced by μ/λ . Imposing $h_0 = 0$, $h_b = 1$,

$$h_i = \begin{cases} \frac{(\mu/\lambda)^i - 1}{(\mu/\lambda)^b - 1}, & \lambda \neq \mu, \\ \frac{i}{b}, & \lambda = \mu. \end{cases} \quad (2.46)$$

The correspondence with the discrete case is exact, and it is the competing clocks of [proposition 2.1](#) that make it so: an embedded jump chain of the birth–death process is a random walk with $p = \lambda/(\lambda + \mu)$, so $q/p = \mu/\lambda$. Mean hitting times satisfy the companion equation $\sum_j q_{ij}(T_j - T_i) = -1$ interior to the targets, and are obtained by the same recursion; they are needed less often here and are not pursued.

2.3.2.5 Extinction probabilities

The ultimate extinction probability of the birth–death process follows from either route above. Letting $t \rightarrow \infty$ in (2.42),

$$\mathbb{P}(\text{extinction}) = \begin{cases} 1, & \lambda \leq \mu, \\ (\mu/\lambda)^N, & \lambda > \mu, \end{cases} \quad (2.47)$$

and the same expression is what (2.46) becomes as $b \rightarrow \infty$ in the supercritical case: the probability of reaching b before 0 tends to one minus the probability of dying first. The threshold — certain extinction at and below criticality, survival with positive probability above — is the continuous-time image of the Galton–Watson threshold of [section 2.2.1.2](#), and [proposition 2.1](#) again supplies the dictionary, since in the parametrisation $p = \lambda/(\lambda + \mu)$ the supercritical condition $\lambda > \mu$ is exactly $p > \frac{1}{2}$, and the survival probability $(\lambda - \mu)/\lambda$ is exactly $(2p - 1)/p$.

2.3.2.6 Conditional means and quasi-stationarity

A process conditioned on an absorbing event not yet having occurred can settle to a time-independent distribution long before absorption; this is *quasi-stationarity*. Formally, a distribution ν on the non-absorbing states of a Markov chain is *quasi-stationary* if $\mathbb{P}(X_t = j \mid X_t \neq 0) = \nu_j$ for all t when $X_0 \sim \nu$, and it is the *Yaglom limit* [7] if the same conditional law is approached from a specified initial state; modern accounts are Méléard and Villemonais [8] and Collet, Martínez and San Martín [9]. For subcritical branching processes satisfying mild non-degeneracy conditions the Yaglom limit exists, and the computations below identify it.

The instrument is the same decomposition in both time domains. For any population process started from N individuals, with $S(t) = \mathbb{P}(X_t > 0)$,

$$\mathbb{E}(X_t \mid X_t > 0) = \frac{\mathbb{E}(X_t)}{S(t)}, \quad (2.48)$$

because the extinct realisations contribute zero. For the subcritical birth–death process, $\mathbb{E}(X_t) = e^{(\lambda - \mu)t}$ and $S(t) = (\lambda - \mu)/(\lambda - \mu e^{(\mu - \lambda)t})$ from (2.42) and (2.43), so

$$\mathbb{E}(X_t \mid X_t > 0) = \frac{\lambda e^{(\lambda - \mu)t} - \mu}{\lambda - \mu} \xrightarrow{t \rightarrow \infty} \frac{\mu}{\mu - \lambda}, \quad (2.49)$$

a constant: the quasi-stationary mean. At criticality the same quotient is $1 + \lambda t$, which grows without bound — there is no stationary conditional law at the critical point.

To compare (2.49) with discrete generations, match the two parametrisations with [proposition 2.1](#): the probability that the birth clock wins the race against death is $p = \lambda/(\lambda + \mu)$, so $\lambda/\mu = p/(1 - p)$. In that parametrisation the quasi-stationary mean is $(1 - p)/(1 - 2p)$; its reciprocal, written

$$A_c(p) = \frac{1 - 2p}{1 - p}, \quad (2.50)$$

is the *continuous-time* quasi-stationary constant. It is an elementary function of p , and that elementary form is the whole reason the continuous-time calculation closes.

The discrete-time counterpart is different, and the difference is the origin of everything in Chapter 3 (*The constant $A(p)$*). For the binary Galton–Watson process with $p < \frac{1}{2}$, (2.48) gives $\mathbb{E}(Z_n \mid Z_n > 0) = (2p)^n/S_n$, and the survival probability satisfies the nonlinear recursion (2.7) rather than a linear equation. The decay is nevertheless geometric at late times,

$$S_n \sim A(p)(2p)^n, \quad (2.51)$$

where

$$A(p) = \lim_{n \rightarrow \infty} \frac{S_n}{(2p)^n}, \quad (2.52)$$

so the conditional mean settles to the constant

$$\mathbb{E}(Z_n \mid Z_n > 0) \xrightarrow{n \rightarrow \infty} \frac{1}{A(p)}, \quad (2.53)$$

the discrete quasi-stationary mean. That the limit exists and that $A(p) \in (0, \frac{1}{2}]$ is established in Chapter 3 (*The constant $A(p)$*) together with the exact product and series representations of the constant, the two-sided bounds pinning it between $\frac{1}{2}A_c(p)$ and $2(1 - 2p)/(3 - 4p)$, the near-critical asymptotic $A(p) \sim 2(1 - 2p)$, and the dynamical obstruction explaining why no elementary formula for it exists. Figure 2.8 shows the convergence and the difficulty near criticality: the closer p lies to $\frac{1}{2}$, the higher the plateau and the longer the approach to it.

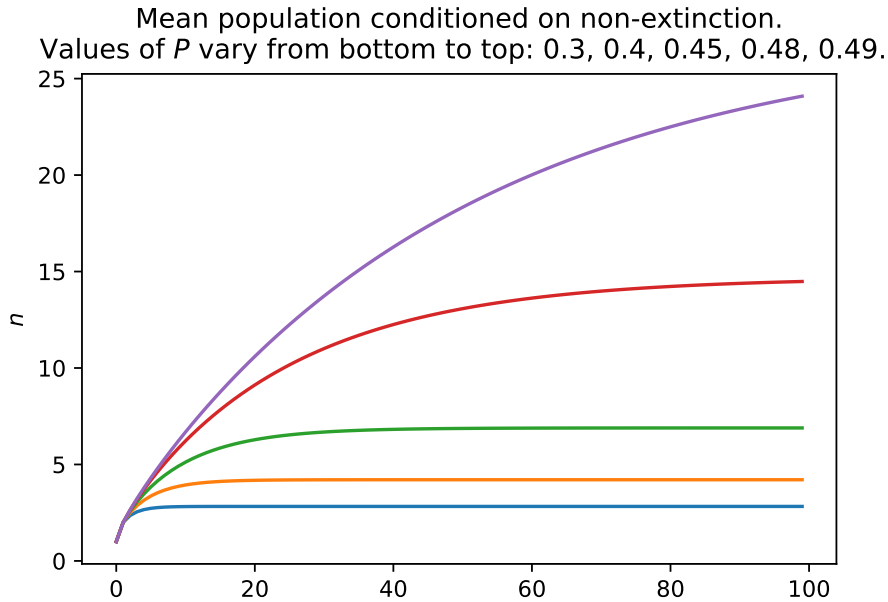


Figure 2.8: The conditional mean $(2p)^n/S_n$ against generation number n , with S_n obtained by iterating (2.7). From bottom to top, $p = 0.3, 0.4, 0.45, 0.48, 0.49$. Each curve approaches the constant $1/A(p)$; the limiting values are 2.825, 4.208, 6.893, 14.703 and 27.476, obtained by iterating the recursion until $S_n/(2p)^n$ stabilises. The lowest three have essentially converged by $n = 100$, but the $p = 0.49$ curve has reached only 24.1 of its limiting 27.5 — the near-critical slowdown analysed in Chapter 3 (*The constant $A(p)$*).

Two further facts round out the quasi-stationary picture without proof. The conditional variance also converges, to a constant depending on p and $A(p)$ alone; the derivation is in [section 2.E](#). And the convergence of the first two moments is consistent with, but does not by itself prove, convergence of the full conditional distribution; that is the content of the Yaglom theorem quoted at the head of this subsection.

2.3.3 The method of characteristics

The generating-function equations assembled so far — (2.18) for the birth–death process, (2.25) for its time-inhomogeneous version, (2.22) for birth–death–catastrophe — all have the same shape: a first-order linear partial differential equation whose coefficients do not involve the unknown. Such equations are solved by the method of characteristics, and this subsection sets the method out in a form that the rest of the thesis can use as a recipe, then works one example through to the end. The remaining members of the family — the absorption-only model, and the absorption–birth–death model whose closed form is an original result of this thesis — are solved in [section 2.G](#) by the identical procedure.

2.3.3.1 The recipe

Consider a first-order equation for G in two state variables and time,

$$\frac{\partial G}{\partial t} = a(x, y, t) \frac{\partial G}{\partial x} + b(x, y, t) \frac{\partial G}{\partial y}, \quad G(x, y, 0) = G_0(x, y), \quad (2.54)$$

which is the form every generating-function equation in this thesis takes once the master equation has been summed. The idea is to find curves $(x(\tau), y(\tau), t(\tau))$ along which G is constant, so that the PDE collapses to an ordinary differential equation — in fact, to nothing at all — on each curve. Introduce a parameter τ along the curves and demand

$$\frac{dG}{d\tau} = 0, \quad \frac{dt}{d\tau} = -1, \quad \frac{dx}{d\tau} = a, \quad \frac{dy}{d\tau} = b, \quad (2.55)$$

with initial data anchored on the surface $t = 0$:

$$x(\sigma, \eta, 0) = \sigma, \quad y(\sigma, \eta, 0) = \eta, \quad t(\sigma, \eta, 0) = 0. \quad (2.56)$$

The choice $dt/d\tau = -1$ makes $\tau = -t$, so characteristics run *backwards* in time from any point (x, y, t) down to the initial surface; that is what allows the initial condition to be read off at the end. The recipe is then:

- (1) solve the characteristic ODEs in (2.55) with the anchoring (2.56);
- (2) since G is constant along each characteristic, $G = G_0(\sigma, \eta)$ everywhere on it;
- (3) eliminate (σ, η, τ) in favour of (x, y, t) by inverting the solutions of step (1);
- (4) substitute into G_0 to obtain $G(x, y, t)$.

Everything in step (3) is algebra; the entire difficulty of any particular problem lives in step (1), where the characteristic ODE for x may be nonlinear. When it is a Riccati equation, as it is for every birth–death-type model in this thesis, it factorises and separates; when the coefficients depend on t , as in (2.25), the same steps go through with t -dependent right-hand sides.

2.3.3.2 Worked example: absorption–death

A compartment takes up particles from a surrounding medium, and particles already inside it may die. Let X_t and Y_t count the particles inside and outside the compartment, started from $X_0 = 0$, $Y_0 = N$. Absorption occurs at rate α per exterior particle and death at rate μ per interior particle; both remove a particle from further consideration, so every particle ends either inside, outside, or dead, and no count can exceed N . The transition probabilities over an infinitesimal interval are

$$\begin{aligned}\mathbb{P}(\Delta X = 1, \Delta Y = -1) &= \alpha j \Delta t, & (\text{absorption}) \\ \mathbb{P}(\Delta X = -1) &= \mu i \Delta t, & (\text{death}) \\ \mathbb{P}(\Delta X = 0, \Delta Y = 0) &= 1 - (\alpha j + \mu i) \Delta t, & (\text{no event})\end{aligned}$$

with forward Kolmogorov equations

$$\frac{dp_{i,j}}{dt} = \alpha(j+1)p_{i-1,j+1} + \mu(i+1)p_{i+1,j} - (\alpha j + \mu i)p_{i,j}. \quad (2.57)$$

Multiplying by $x^i y^j$, summing, and shifting indices as in [section 2.3.1.1](#) gives the generating-function equation, in exactly the form [\(2.54\)](#):

$$\frac{\partial G}{\partial t} = \mu(1-x)\frac{\partial G}{\partial x} + \alpha(x-y)\frac{\partial G}{\partial y}, \quad G(x, y, 0) = y^N. \quad (2.58)$$

Step 1: the characteristic ODEs. From [\(2.55\)](#) with $a = \mu(1-x)$ and $b = \alpha(x-y)$,

$$\frac{dG}{d\tau} = 0, \quad (2.59)$$

$$\frac{dt}{d\tau} = -1, \quad (2.60)$$

$$\frac{dx}{d\tau} = \mu(1-x), \quad (2.61)$$

$$\frac{dy}{d\tau} = \alpha(x-y). \quad (2.62)$$

[\(2.60\)](#) gives $t = -\tau$ at once. [\(2.61\)](#) separates: $1-x = (1-\sigma)e^{-\mu\tau}$. Substituting that into [\(2.62\)](#) and solving with the integrating factor $e^{\alpha\tau}$,

$$y = 1 - \frac{\alpha}{\alpha - \mu}(1 - \sigma)e^{-\mu\tau} + c(\sigma, \eta)e^{-\alpha\tau},$$

and the anchoring [\(2.56\)](#) fixes $c(\sigma, \eta) = \eta + \frac{\alpha}{\alpha - \mu}(1 - \sigma) - 1$, so

$$y = 1 - \frac{\alpha}{\alpha - \mu}(1 - \sigma)e^{-\mu\tau} + \left(\eta + \frac{\alpha}{\alpha - \mu}(1 - \sigma) - 1 \right) e^{-\alpha\tau}. \quad (2.63)$$

Steps 2–3: constant on characteristics, then inversion. By [\(2.59\)](#), G equals its initial value on each characteristic, and under [\(2.56\)](#) the initial condition reads $G_0(\sigma, \eta) = \eta^N$:

$$G = \eta^N. \quad (2.64)$$

It remains to express η in terms of (x, y, t) . Rearranging [\(2.63\)](#) for η , substituting $1 - \sigma = (1 - x)e^{\mu\tau}$ from the x -solution and $\tau = -t$, gives after simplification

$$\eta = 1 - \frac{\mu}{\mu - \alpha}e^{-\alpha t} + \frac{\alpha}{\mu - \alpha}e^{-\mu t} + \frac{\alpha}{\mu - \alpha}(e^{-\alpha t} - e^{-\mu t})x + e^{-\alpha t}y.$$

Step 4: the generating function. Substituting into (2.64),

$$G(x, y, t) = \left(1 - \frac{\mu}{\mu - \alpha} e^{-\alpha t} + \frac{\alpha}{\mu - \alpha} e^{-\mu t} + \frac{\alpha}{\mu - \alpha} (e^{-\alpha t} - e^{-\mu t}) x + e^{-\alpha t} y \right)^N, \quad (2.65)$$

with the removable degeneracy $\alpha = \mu$ understood by its limit,

$$\frac{\alpha}{\mu - \alpha} (e^{-\alpha t} - e^{-\mu t}) \xrightarrow{\mu \rightarrow \alpha} \alpha t e^{-\alpha t}. \quad (2.66)$$

The check. Because particles neither interact nor multiply in this model, the answer can be obtained independently, and the two routes should agree. A single particle has state probabilities

$$p_{0,1}(t) = e^{-\alpha t}, \quad p_{1,0}(t) = \frac{\alpha}{\mu - \alpha} (e^{-\alpha t} - e^{-\mu t}), \quad p_{0,0}(t) = 1 - p_{0,1}(t) - p_{1,0}(t), \quad (2.67)$$

outside, inside and dead respectively (figure 2.9). The N particles act independently, so the joint generating function is the N th power of the single-particle one — which is precisely the form (2.65), with the three bracket terms as the probabilities of dead, inside and outside. Setting $x = y = 1$ confirms that the three probabilities sum to one, and setting $x = 1$ alone gives $G(1, y, t) = (e^{-\alpha t} y + 1 - e^{-\alpha t})^N$: the exterior count is binomial, as it must be, since exterior particles are absorbed independently at rate α regardless of what happens inside.

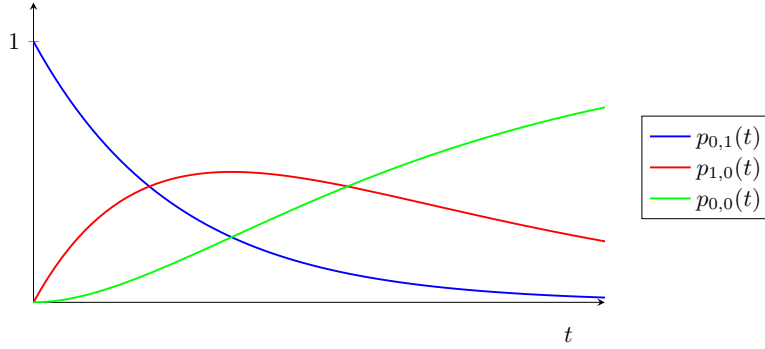


Figure 2.9: State probabilities for a single particle in the absorption–death model. The particle starts outside (blue, decaying at rate α), is absorbed into the compartment (red, which rises and then decays as death takes over), and finally dies (green, absorbing). The three curves sum to one at every time, and the coefficients of (2.65) are read off them.

The same four steps, applied to harder coefficients, produce the remaining closed forms of this thesis. Adding interior birth to the model above opens an infinite state space, the independence check fails, and the characteristic calculation becomes the only route; it is carried out in section 2.G, whose end product — the absorption–birth–death generating function, (2.126) — is the closed form on which the absorption models of later chapters are built. The two support appendices belong to this subsection as well: section 2.A records the hypergeometric integral identity that the harder calculation rests on, and section 2.B explains how to recover time-dependent state probabilities from a generating function given by a formula rather than a series.

2.A An integral identity for the hypergeometric function

The route to (2.124) taken in section 2.G.2 expands the characteristic $x(\tau)$ in a geometric series and integrates term by term. An alternative is to integrate the quotient (2.121) directly, splitting

the numerator into two pieces, each of the form

$$I(\tau) = \int \frac{e^{h\tau}}{P + Qe^{k\tau}} d\tau \quad (2.68)$$

with h, k, P, Q constant. That route is longer and easier to get wrong — the two pieces carry different prefactors, and dropping one produces a formula that still satisfies the initial condition while violating $G(1, 1, t) = 1$ — but the identity it rests on is useful in its own right, so it is recorded and proved here.

Proposition 2.4. *For $h/k > 0$ and $-\frac{Q}{P}e^{k\tau} \notin [1, \infty)$,*

$$\int \frac{e^{h\tau}}{P + Qe^{k\tau}} d\tau = \frac{e^{h\tau}}{Ph} {}_2F_1\left(1, \frac{h}{k}; \frac{h}{k} + 1; -\frac{Q}{P}e^{k\tau}\right) + \text{const}, \quad (2.69)$$

where the hypergeometric function is continued from its principal power series along the interval of integration.

Proof. We use the Euler integral representation of the hypergeometric function [10],

$${}_2F_1(a, b; c; z) = \frac{\Gamma(c)}{\Gamma(b)\Gamma(c-b)} \int_0^1 u^{b-1}(1-u)^{c-b-1}(1-zu)^{-a} du, \quad (2.70)$$

Γ being the gamma function.

Start from (2.68) and substitute $v = e^{k\tau}$, so that $d\tau = k^{-1}v^{-1}dv$:

$$I = \frac{1}{k} \int \frac{v^{h/k}}{P + Qv} v^{-1} dv = \frac{1}{Pk} \int \frac{v^{h/k-1}}{1 + \frac{Q}{P}v} dv.$$

Fix the antiderivative by writing this as a definite integral from 0 to v in a dummy variable w ; the integral converges at the lower limit precisely because $h/k > 0$:

$$I = \frac{1}{Pk} \int_0^v \frac{w^{h/k-1}}{1 + \frac{Q}{P}w} dw.$$

Now rescale, $w = vu$, so that $dw = v du$ with v held fixed:

$$I = \frac{1}{Pk} \int_0^1 \frac{(vu)^{h/k-1}}{1 + \frac{Q}{P}vu} v du = \frac{v^{h/k}}{Pk} \int_0^1 \frac{u^{h/k-1}}{1 + \frac{Q}{P}vu} du. \quad (2.71)$$

The remaining integrand matches that of (2.70) with $b = h/k$, $c = h/k + 1$, $a = 1$ and $z = -\frac{Q}{P}v$, since then $(1-u)^{c-b-1} = 1$ and $(1-zu)^{-a} = (1 + \frac{Q}{P}vu)^{-1}$. Hence, using $\Gamma(z+1) = z\Gamma(z)$,

$$\int_0^1 \frac{u^{h/k-1}}{1 + \frac{Q}{P}u} du = \frac{\Gamma(\frac{h}{k})\Gamma(1)}{\Gamma(\frac{h}{k}+1)} {}_2F_1\left(1, \frac{h}{k}; \frac{h}{k} + 1; -\frac{Q}{P}v\right) = \frac{k}{h} {}_2F_1\left(1, \frac{h}{k}; \frac{h}{k} + 1; -\frac{Q}{P}v\right).$$

Substituting into (2.71) gives

$$I = \frac{v^{h/k}}{Ph} {}_2F_1\left(1, \frac{h}{k}; \frac{h}{k} + 1; -\frac{Q}{P}v\right),$$

and restoring $v = e^{k\tau}$ gives (2.69) on the Euler domain. Both sides are analytic antiderivatives away from the singular set, so continuation along any admissible interval preserves the identity. $\square$

Remark 2.5. Applied to the two pieces of (2.121) — the first with $h = \alpha$, the second with $h = \alpha + b_1$, and both with $k = b_1$, $P = \sigma - x_-$ and $Q = -(\sigma - x_+)$ — proposition 2.4 reproduces (2.136) once the prefactors $\alpha x_+(\sigma - x_-)$ and $-\alpha x_-(\sigma - x_+)$ are carried through correctly. The factor $(\sigma - x_+)/(\sigma - x_-)$ attached to the second piece is easily lost, and its loss is invisible at $t = 0$; checking $G(1, 1, t) = 1$ catches it immediately. At the resonant values $\alpha = n(\mu - \lambda)$ of remark 2.8 the same conclusion follows by the finite parameter limit described there.

2.B Extracting state probabilities from a generating function

A generating function defined by a formula rather than as an explicit series — for example

$$G(x, y, t) = \left((1 - e^{-\alpha t})x + e^{-\alpha t}y \right)^n \quad (2.72)$$

— can always be converted back to the form

$$G(x, y, t) = \sum_{i=0}^{\infty} \sum_{j=0}^{\infty} p_{i,j}(t) x^i y^j, \quad (2.73)$$

recovering the time-dependent state probabilities. All that is required is that G be repeatedly partially differentiable in x and y .

The reason is the bivariate Maclaurin expansion,

$$f(x, y) = \sum_{k=0}^{\infty} \frac{1}{k!} \sum_{\varrho=0}^k \binom{k}{\varrho} \partial_x^{\varrho} \partial_y^{k-\varrho} f \Big|_{(0,0)} x^{\varrho} y^{k-\varrho},$$

whose coefficient of $x^i y^j$, taking $k = i + j$ and $\varrho = i$, is $\frac{1}{(i+j)!} \binom{i+j}{i} \partial_x^i \partial_y^j f \Big|_{(0,0)}$. Hence

$$p_{i,j}(t) = \frac{1}{i! j!} \partial_x^i \partial_y^j G \Big|_{(0,0)}. \quad (2.74)$$

For completeness, the coefficient is also represented by the bivariate Cauchy integral

$$p_{i,j}(t) = \frac{1}{(2\pi i)^2} \int_{|x|=\varrho_x} \int_{|y|=\varrho_y} \frac{G(x, y, t)}{x^{i+1} y^{j+1}} dy dx, \quad (2.75)$$

for circles inside the domain of analyticity; this form is useful for stable numerical extraction when derivatives of a special-function representation are unwieldy.

The multilinear case

Both (2.110) and (2.65) have the form

$$G(x, y, t) = (c(t) + f(t)x + g(t)y)^n, \quad (2.76)$$

for which the derivatives are immediate. Each differentiation lowers the power by one and brings down a factor, so

$$\partial_x^a \partial_y^b G = \frac{n!}{(n-a-b)!} f^a g^b (c + fx + gy)^{n-a-b} \quad (a + b \leq n), \quad (2.77)$$

and zero for $a + b > n$. Evaluating at the origin and applying (2.74),

$$p_{i,j}(t) = \frac{n!}{i!j!(n-i-j)!} f(t)^i g(t)^j c(t)^{n-i-j}, \quad i + j \leq n, \quad (2.78)$$

and $p_{i,j} = 0$ otherwise. This is the trinomial distribution, which is exactly what it should be: the n particles are independent, and at time t each is in one of three states, with probabilities c (neither inside nor outside — that is, dead), f (inside) and g (outside). Note in particular that the falling factorial $n!/(n-i-j)!$ in (2.77) is *not* n^{i+j} ; the two agree only to leading order in n , and using the latter breaks normalisation.

Applied to the absorption models

For absorption only, (2.72), we have $c = 0$, $f = 1 - e^{-\alpha t}$ and $g = e^{-\alpha t}$. The factor c^{n-i-j} then vanishes unless $i + j = n$, and (2.78) gives

$$p_{i,j}(t) = \binom{n}{i} (1 - e^{-\alpha t})^i (e^{-\alpha t})^j, \quad i + j = n,$$

so $X_t \sim \text{Binomial}(n, 1 - e^{-\alpha t})$ and $Y_t = n - X_t$, as anticipated in section 2.G.1.

For absorption–death, (2.65), the three functions are

$$c(t) = 1 - \frac{\mu}{\mu - \alpha} e^{-\alpha t} + \frac{\alpha}{\mu - \alpha} e^{-\mu t}, \quad f(t) = \frac{\alpha}{\mu - \alpha} (e^{-\alpha t} - e^{-\mu t}), \quad g(t) = e^{-\alpha t},$$

which are precisely $p_{0,0}$, $p_{1,0}$ and $p_{0,1}$ of (2.67), with the continuous equal-rate limit of (2.66) when $\alpha = \mu$, so that

$$p_{i,j}(t) = \frac{n!}{i!j!(n-i-j)!} f(t)^i g(t)^j c(t)^{n-i-j}.$$

Each particle independently ends up dead, absorbed, or still outside, with those three probabilities, and the joint law is trinomial; since $c + f + g = 1$, summing over i and j recovers one by the multinomial theorem. That the generating function factorises in this way is (2.109) read backwards.

The absorption–birth–death model does not have the multilinear form (2.76), and its coefficient extraction — which separates a y -binomial factor from a univariate convolution — is recorded with the model itself, in section 2.G.3.

2.C Critical branching in detail

This appendix develops the critical theory of the binary Galton–Watson process that section 2.2.1.2 summarises: the certainty of extinction at and below criticality, the power-law survival and infinite expected lifetime at criticality, and the total-progeny exponent. The asymptotics established here are used again in the near-critical analysis of Chapter 3 (*The constant $A(p)$*).

Certainty of extinction

Since S_n is non-increasing and bounded below by 0, it converges, and its limit must be a fixed point of (2.7). Setting $S_{n+1} = S_n = S_\infty$,

$$S_\infty = 2pS_\infty - pS_\infty^2 \quad \Longleftrightarrow \quad pS_\infty \left(S_\infty + \frac{1}{p} - 2 \right) = 0,$$

with roots

$$S_\infty = 0 \quad \text{and} \quad S_\infty = 2 - \frac{1}{p} = \frac{2p-1}{p}.$$

For $p < \frac{1}{2}$ the second root is negative and carries no probabilistic meaning, so the only admissible limit is $S_\infty = 0$ and extinction is certain. At $p = \frac{1}{2}$ the two roots collide at the origin and extinction remains certain. For $p > \frac{1}{2}$ a positive root appears, and it is that root which is attained: the process survives forever with positive probability $S_\infty = (2p-1)/p$. Figure 2.10 shows the transition, together with the change in stability of the origin that produces it.

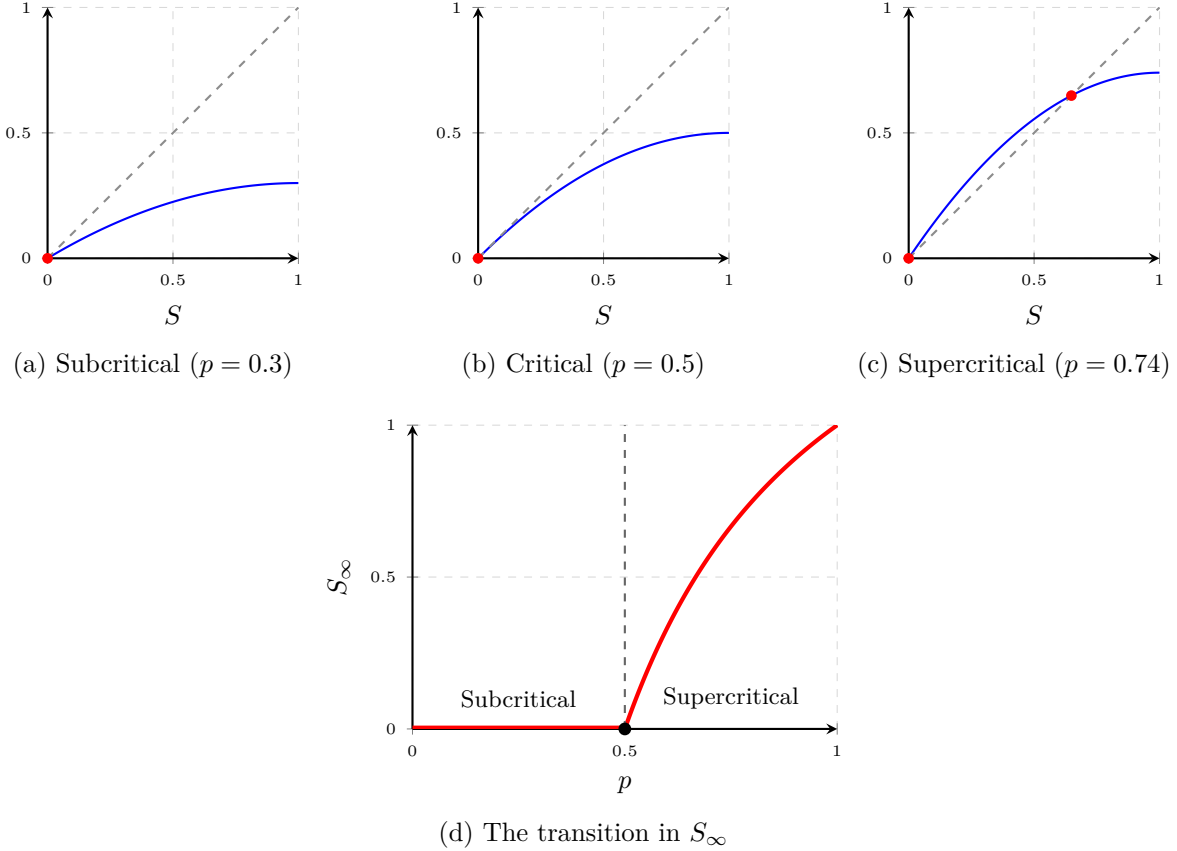


Figure 2.10: Fixed points of the survival recursion (2.7). Panels (a)–(c) show the map $S \mapsto 2pS - pS^2$ (blue) against the diagonal (dashed), with the fixed points marked in red. Below and at criticality the origin is the only fixed point in $[0, 1]$ and it is attracting, so extinction is certain; at $p = \frac{1}{2}$ the map is tangent to the diagonal there. For $p > \frac{1}{2}$ the origin turns repelling and a stable positive fixed point $S_\infty = (2p-1)/p$ separates from it, traced in panel (d).

The critical case: a power law and an infinite expected lifetime

At $p = \frac{1}{2}$ the expectation is constant, $\mathbb{E}(Z_n) = 1$ for every n , and yet extinction is certain. The two facts are reconciled by the tail: rare realisations survive for enormously long times and grow enormously large, and they do so at exactly the rate needed to hold the mean at one.

Let

$$T = \inf\{n : Z_n = 0\}$$

be the extinction time, so that the process passes through generations $0, 1, \dots, T-1$ and is dead at T . Writing T as a sum of indicators and taking expectations term by term — legitimate because every term is non-negative —

$$\mathbb{E}(T) = \sum_{n=0}^{\infty} \mathbb{P}(T > n) = \sum_{n=0}^{\infty} \mathbb{P}(Z_n > 0) = \sum_{n=0}^{\infty} S_n. \quad (2.79)$$

The question is therefore how fast S_n decays at criticality.

Setting $p = \frac{1}{2}$ in (2.7) gives $S_{n+1} - S_n = -\frac{1}{2}S_n^2$. Once n is large the unit generational increment is small compared with the scale on which S_n varies, and the difference on the left is well approximated by a derivative. This step is heuristic — it replaces a difference equation by a differential one, and nothing here justifies the replacement beyond the smallness of the increment relative to the solution — but its conclusion is standard and is confirmed by simulation below. With that caveat,

$$\frac{dS}{dn} = -\frac{1}{2}S^2, \quad \text{with general solution} \quad S(n) = \frac{2}{n + \text{const}},$$

so that

$$S_n \sim \frac{2}{n} \quad (n \rightarrow \infty), \quad (2.80)$$

as figure 2.12(a) confirms. The same asymptotic can in fact be obtained directly from the exact recursion, without a continuum approximation: at $p = \frac{1}{2}$ the recursion reads $S_{n+1} = S_n(1 - S_n/2)$, and since $S_n \downarrow 0$ the reciprocal sequence has increments

$$\frac{1}{S_{n+1}} - \frac{1}{S_n} = \frac{1}{2(1 - S_n/2)} \rightarrow \frac{1}{2}. \quad (2.81)$$

Summing the convergent increments — equivalently, applying the Stolz–Cesàro theorem — gives $(1/S_n)/n \rightarrow \frac{1}{2}$, which is precisely (2.80). This rigorous route, and the increment (2.81), are used again in the near-critical analysis of Chapter 3 (*The constant $A(p)$*).

Substituting into (2.79), the tail of the sum behaves like twice the harmonic series,

$$\mathbb{E}(T) = \sum_{n=0}^{\infty} S_n \sim \sum_{n=1}^{\infty} \frac{2}{n},$$

which diverges (figure 2.11). Since every S_n is positive, $\mathbb{E}(T) = \infty$: the expected lifetime of the critical Galton–Watson process is infinite even though extinction occurs with probability one. The point probabilities of the lifetime follow from the exact decrement of the recursion:

$$\mathbb{P}(T = n + 1) = S_n - S_{n+1} = \frac{1}{2}S_n^2 \sim \frac{2}{n^2}, \quad (2.82)$$

the heavy tail visible in figure 2.12(b).

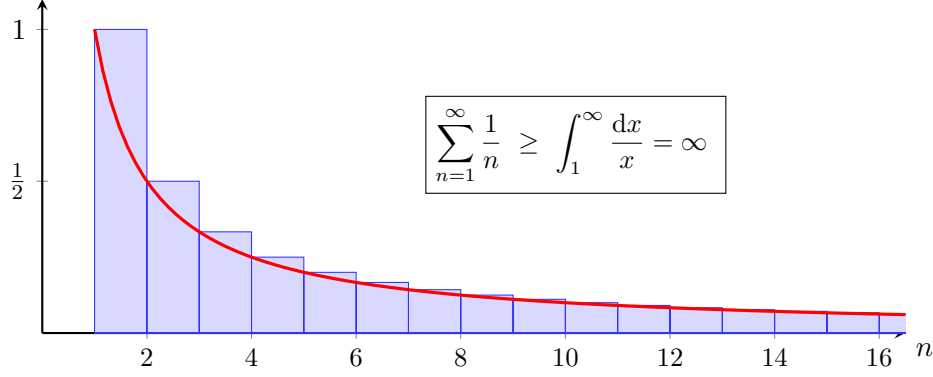


Figure 2.11: Divergence of the harmonic series. Each blue rectangle has area $1/n$ and contains the region under the red curve $1/x$ on $[n, n+1]$, so the sum of the areas exceeds $\int_1^\infty x^{-1} dx = \infty$. Because $S_n \sim 2/n$ at criticality, the same divergence carries over to $\mathbb{E}(T) = \sum_n S_n$.

A third critical exponent belongs to the total progeny

$$\mathcal{T} = \sum_{n \geq 0} Z_n,$$

and it is worth recording because [figure 2.12\(c\)](#) displays it. A finite binary genealogy with k division vertices has $k+1$ terminal vertices and hence $2k+1$ individuals in all, and the number of such ordered full binary trees is the Catalan number $C_k = (k+1)^{-1} \binom{2k}{k}$. Each division contributes a factor p and each death a factor $1-p$, so the exact law is

$$\mathbb{P}(\mathcal{T} = 2k+1) = C_k p^k (1-p)^{k+1}, \quad (2.83)$$

zero at even values. At criticality Stirling's formula gives

$$\mathbb{P}(\mathcal{T} = 2k+1) \sim \frac{1}{2\sqrt{\pi}} k^{-3/2}, \quad (2.84)$$

the classical $3/2$ law of critical branching, here along its admissible odd support.

The heavy tail responsible for all of these exponents is visible directly in simulation. A power-law distributed quantity carries a modest but non-negligible chance of an extraordinarily large realisation, and a sample mean computed from such a quantity does not settle: the larger the sample, the larger the mean, because the occasional enormous value keeps pushing the normalised sum upward. [Figure 2.12](#) collects the three classical critical exponents — for the survival probability, for the extinction time and for the total progeny.

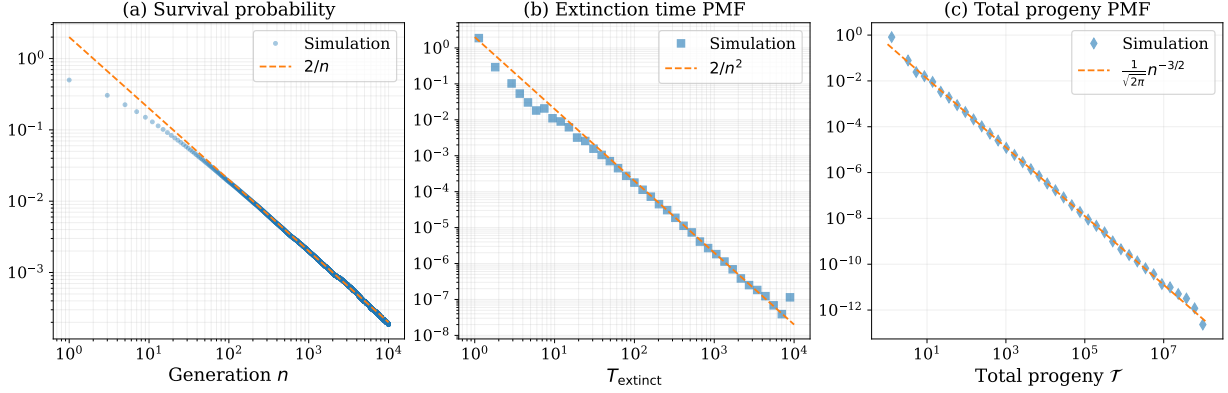


Figure 2.12: Power-law behaviour in the critical Galton–Watson process ($p = 0.5$), from 10^6 realisations. (a) Survival probability $S_n = \mathbb{P}(Z_n > 0)$, showing the late-generation scaling $S_n \sim 2/n$ of (2.80). (b) Probability mass function of the extinction time $T = \inf\{n : Z_n = 0\}$, with heavy tail $\mathbb{P}(T = n) \sim 2/n^2$ as in (2.82). (c) Probability mass function of the total progeny $\mathcal{T} = \sum_{k \geq 0} Z_k$, displaying the classical critical branching asymptotic $\mathbb{P}(\mathcal{T} = n) \sim n^{-3/2}$ of (2.84).

2.D Founding cohorts and early survival

This appendix collects the early-generation calculations for the binary Galton–Watson process that later modelling chapters draw on: the survival probability of a founding cohort of size k , the arithmetic of the first few generations near criticality, and the conditioning effect known as the push of the past.

The first generations near criticality

Return to the binary process of section 2.2.1.2 near critical, either just below or just above. With the probability of division p and of death $1 - p$ each close to a half, a lineage dies at its first round almost half the time. Should it survive one generation, both its offspring fail at their own first attempt with probability around 0.25, so the lineage reaches the second generation only with probability about 0.375:

$$0.375 \approx \underbrace{(1 - 0.5)}_{\text{no extinction at the first step}} \times \underbrace{(1 - 0.5 \times 0.5)}_{\text{nor at the second}}.$$

More generally, survival to a given generation is obtained by iterating the probability generating function. With $\phi(z) = pz^2 + (1 - p)$ from (2.6), the extinction and survival probabilities follow from

$$u_{n+1} = pu_n^2 + 1 - p, \quad S_{n+1} = 2pS_n - pS_n^2, \quad (2.85)$$

started at $u_0 = 0$, $S_0 = 1$. At exactly critical branching this gives, for $n = 1, 2, 3, 4$,

$$u_n = 0.500, 0.625, 0.695, 0.741 \quad (3 \text{ s.f.}),$$

so the probability of survival decays quickly over the first few rounds of division, and does so almost as fast just above criticality as just below. Figure 2.13 makes the point: whatever advantage supercriticality confers, it is not an early-generation advantage.

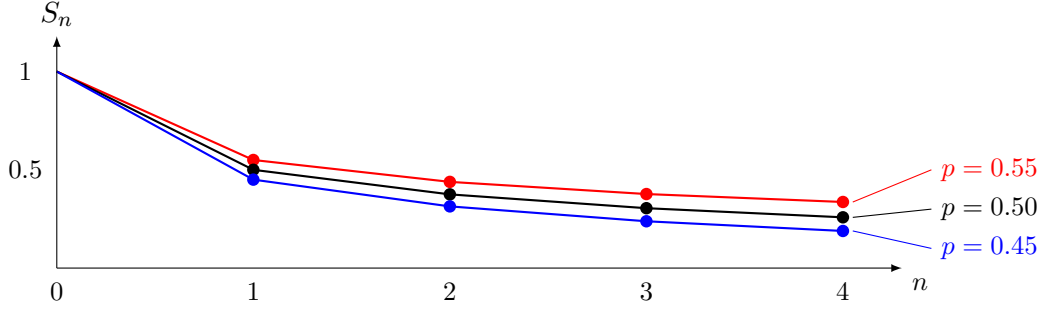


Figure 2.13: Survival probability over the first four generations from a single progenitor, computed from (2.85). Even in the supercritical case ($p = 0.55$) the early decay is steep, and the three curves have barely separated by $n = 4$.

An initial cohort of size k

Now suppose $Z_0 = k$: the process begins not with a single individual but with a group, small to moderate in number, perhaps $k = 10$. Under supercritical branching, growth becomes more predictably exponential as the count rises, because a large number of individuals lets the expected dynamics show through the noise of individual events. The founding cohort is precisely the regime in which that noise dominates instead.

What, then, are such a cohort's chances of long-run persistence? The question is answered by asking for the probability that at least one descendant remains, that is, that not every one of the k founding lineages has died out. The lineages are independent and each is extinct by generation n with probability u_n , so all k have gone with probability u_n^k , and

$$S_n^{(k)} = 1 - u_n^k = 1 - (1 - S_n)^k. \quad (2.86)$$

The identity separates two effects: S_n contains the dynamics of one lineage, whereas the exponent k contains the founding size. For small S_n one has $S_n^{(k)} \sim kS_n$ when $kS_n \ll 1$, but the exact formula is needed once the survival chance is no longer small.

Figures 2.14 and 2.15 show $S_n^{(k)}$ through 500 generations for $k = 1, 10, 100, 1000$. The effect visible in figure 2.14 — that a larger first cohort gives improved longevity — is highly sensitive to criticality. Persistence improves quickly as p approaches $\frac{1}{2}$ from below and continues to improve dramatically into supercriticality, and the two panels of figure 2.15 differ only in the third decimal place of p .

Conditioning and apparent early growth

The conditioning of section 2.3.2.6 has a mirror image which is worth recording, because it shows that the effect is not confined to the subcritical case.

Consider an ensemble of independent homogeneous birth–death processes, all with the same constant rates, and retain only those that happen still to be alive at some large time. The retained subpopulation is automatically enriched for early rapid growth: a realisation that grew quickly at the outset is far more likely to have escaped the early extinction risk documented in figure 2.13, and so far more likely to be in the retained set. No change in the underlying rates is required to produce the enrichment, and the apparent slowdown that follows is not a slowdown at all but a reversion to the mean. The effect is known as the *push of the past*, and was identified in a setting far from the present one, as an explanation for a systematic pattern in longevity and diversity data [11].

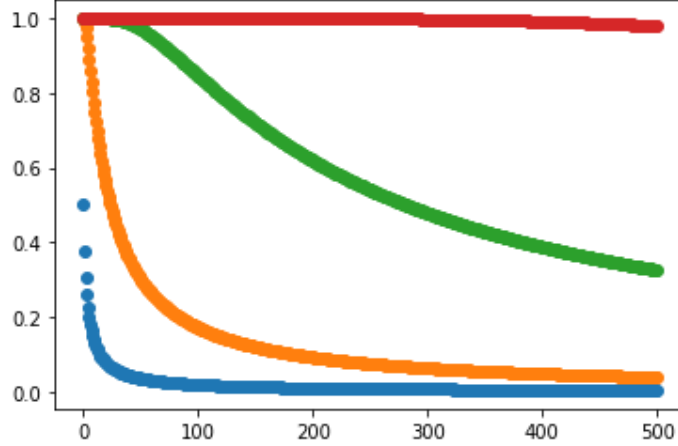


Figure 2.14: Survival probability $S_n^{(k)}$ from (2.86) out to 500 generations at exact criticality, $p = 0.5$, for $k = 1, 10, 100, 1000$ (blue through to red). A larger founding cohort buys longevity, but at criticality every curve is still headed for zero.

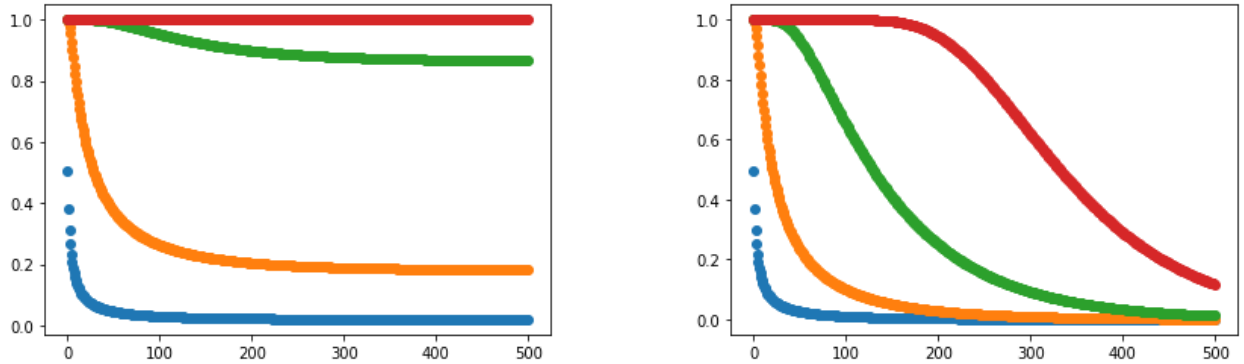


Figure 2.15: As figure 2.14, but with p displaced slightly from criticality: $p = 0.505$ on the left (supercritical), $p = 0.495$ on the right (subcritical). The benefit of a large initial cohort is acutely sensitive to which side of $\frac{1}{2}$ the process sits on. A displacement of half a per cent separates curves that were nearly coincident in figure 2.14.

That is exactly the conditioning of section 2.3.2.6 seen from the other side. There a subcritical process was conditioned on survival and its mean settled to $1/A(p)$ rather than decaying; here a critical or supercritical process is conditioned on survival to a late time and its early growth appears inflated. In both cases the conditioning, not the mechanism, does the work: selection is being performed on paths after they have been generated.

The parallel suggests a modelling hypothesis that later chapters take up: that early replication weighs heavily on eventual survival, even when later per-capita rates are unchanged. Figure 2.13 is the reason to take it seriously, since the first few generations are where nearly all of the extinction risk sits.

2.E Quasi-stationary variance of the binary Galton–Watson process

section 2.3.2.6 establishes that the subcritical binary Galton–Watson process, conditioned on survival, has limiting mean $1/A(p)$. The mean alone does not establish quasi-stationarity: to show that the conditional *distribution* settles we need at least the second moment, and this appendix computes it.

The second moment

For a random variable X the moment generating function is $\mathbb{E}(e^{tX})$, which in terms of the probability generating function is $\phi(e^t)$. Write $M_n(t) = \phi_n(e^t)$ for the moment generating function of Z_n , so that moments are recovered by

$$\mathbb{E}(Z_n^\rho) = \left. \frac{d^\rho}{dt^\rho} M_n(t) \right|_{t=0}. \quad (2.87)$$

For the binary offspring law (2.6) the first-generation function is

$$M_1(t) = \phi(e^t) = p e^{2t} + (1 - p), \quad (2.88)$$

whose second derivative at $t = 0$ is $4p$, so $\mathbb{E}(Z_1^2) = 4p$. Reaching generation n requires a little more. Since ϕ_n is the n -fold composition of ϕ ,

$$M_n(t) = \phi(\phi(\dots \phi(e^t) \dots)),$$

and differentiating by the chain rule gives a product of $n + 1$ factors,

$$M'_n(t) = \phi'(\phi_{n-1}(e^t)) \cdot \phi'(\phi_{n-2}(e^t)) \cdots \phi'(\phi_1(e^t)) \cdot \phi'(e^t) \cdot e^t. \quad (2.89)$$

Now $\phi'(z) = 2pz$ for the binary law, so each factor collapses,

$$\phi'(\phi_k(e^t)) = 2p \phi_k(e^t) = 2p M_k(t), \quad \phi'(e^t) = 2p e^t,$$

and (2.89) becomes

$$M'_n(t) = (2p)^n M_{n-1}(t) M_{n-2}(t) \cdots M_1(t) e^{2t}. \quad (2.90)$$

Before differentiating again, note that every factor on the right equals one at $t = 0$: $M_k(0) = 1$ for all k , and $e^0 = 1$. That is what makes the second derivative tractable. Differentiating the product gives a sum of n terms, each identical to (2.90) except that a single factor has been differentiated, and at $t = 0$ all the undifferentiated factors become one, leaving

$$M''_n(0) = (2p)^n [M'_{n-1}(0) + M'_{n-2}(0) + \cdots + M'_1(0) + 2].$$

But $M'_k(0)$ is the mean population at generation k , namely $(2p)^k$, so

$$M''_n(0) = (2p)^n [(2p)^{n-1} + \cdots + (2p) + 2] = (2p)^n \left[\sum_{i=0}^{n-1} (2p)^i + 1 \right].$$

Summing the finite geometric series and rearranging,

$$\mathbb{E}(Z_n^2) = (2p)^n \left(1 + \frac{1}{1 - 2p} \right) - (2p)^{2n} \left(\frac{1}{1 - 2p} \right). \quad (2.91)$$

The minus sign is not optional: at $n = 1$ the expression must return $\mathbb{E}(Z_1^2) = 4p$, and it does.

Late-time variance

From $\text{Var}(Z_n) = \mathbb{E}(Z_n^2) - \mathbb{E}(Z_n)^2$ and (2.91),

$$\text{Var}(Z_n) = (2p)^n \left(1 + \frac{1}{1-2p}\right) - (2p)^{2n} \left(\frac{1}{1-2p}\right) - (2p)^{2n},$$

and for $2p < 1$ the terms in $(2p)^{2n}$ are negligible beside $(2p)^n$ at large n , so $\text{Var}(Z_n) \rightarrow (2p)^n(1 + 1/(1-2p))$. It is tempting to read stationarity of the conditional variance straight off that, but the step is not available: the conditional variance is not the unconditional one divided by anything simple. Start instead from the conditional moments, using (2.51) in each:

$$\begin{aligned} \mathbb{E}(Z_n^2 \mid Z_n > 0) &= \frac{\mathbb{E}(Z_n^2)}{S_n} \xrightarrow{n \rightarrow \infty} \frac{(2p)^n \left(1 + \frac{1}{1-2p}\right)}{A(p)(2p)^n} = \frac{1}{A(p)} \left(1 + \frac{1}{1-2p}\right), \\ \mathbb{E}(Z_n \mid Z_n > 0) &= \frac{\mathbb{E}(Z_n)}{S_n} \xrightarrow{n \rightarrow \infty} \frac{1}{A(p)}. \end{aligned}$$

Both limits are finite, so the conditional variance tends to a constant:

$$\text{Var}(Z_n \mid Z_n > 0) \rightarrow \frac{1}{A(p)} \left(1 + \frac{1}{1-2p}\right) - \frac{1}{A(p)^2}. \quad (2.92)$$

The first two moments of the conditional population therefore both stabilise, and the Yaglom theorem quoted in section 2.3.2.6 supplies the full quasi-stationary distribution whose first moments these are. Note that (2.92), like (2.53), is expressed entirely in terms of p and the single unknown constant $A(p)$, whose analysis is the subject of Chapter 3 (*The constant $A(p)$*).

2.F Discrete birth–death–catastrophe

This appendix develops the discrete-generation counterpart of the birth–death–catastrophe process of definition 2.2: exact probabilities for rupture and survival in discrete time, the weighted generating-function recursion that carries the joint law, and the killed-generator formulation of quasi-stationarity used by the continuous-time analysis.

Probability of catastrophe at step n

Let κ be the probability that a given individual initiates catastrophe at a step, and write $c = 1 - \kappa$. A single individual fails to do so with probability c , two with c^2 , and a cohort of size N with c^N .

Consider first birth and catastrophe alone, with no death, so $p = 1$. Then the census of the living population at any generation is known with certainty: from a single progenitor the population simply doubles at each step prior to catastrophe. Conditional on reaching step n , where 2^{n-1} individuals are exposed, the process therefore comes to its abrupt halt at that step with probability

$$h_n = 1 - c^{2^{n-1}}, \quad (2.93)$$

one minus the probability that none of the 2^{n-1} individuals present caused it. This is a *conditional* hazard: it assumes step n is reached. The unconditional probability that catastrophe first occurs at step n is smaller, because every earlier step must also have been passed.

Probability of process survival

Still with no death, define S_n as the probability that the process survives to generation n . Survival to generation 0 is certain, $S_0 = 1$. Survival to generation 1 requires only that the founder divided without catastrophe, $S_1 = c$. Persistence to step 2 requires that the founder divided and that each of its two offspring did too, so $S_2 = c^2$. The pattern is clear: survival to generation n requires every prior step to have been passed, and therefore

$$S_n = \prod_{i=0}^{n-1} c^{2^i} = c^{\sum_{i=0}^{n-1} 2^i} = c^{2^n - 1}. \quad (2.94)$$

The unconditional mass function of the first-catastrophe time follows at once:

$$\mathbb{P}([K = n] | \text{no catastrophe before } n) = S_n h_n = c^{2^n - 1} (1 - c^{2^n}). \quad (2.95)$$

The factor S_n is indispensable; h_n alone is a conditional hazard, not an exact-time probability, and conflating the two is the standard error of mean-field hazard reasoning, which substitutes the current expected census into a rate as though the census were deterministic.

Death included

The difficulty comes once death is allowed, because the size of the living population is then no longer known with certainty from one generation to the next. Let $\tilde{Z}_i$ denote the underlying binary Galton–Watson population with the catastrophe mechanism switched off, and write

$$H_n = \sum_{i=0}^{n-1} \tilde{Z}_i \quad (2.96)$$

for the cumulative exposure over the first n steps. Conditioning on the whole catastrophe-free genealogy gives the exact identity

$$\mathbb{P}([K = n] | \text{no catastrophe before } n) = \mathbb{E}(c^{H_n}). \quad (2.97)$$

The natural guess is that the probabilities shake out when the expected population is used in place of the actual one:

$$\mathbb{E}(c^{H_n}) \stackrel{?}{=} c^{\sum_{i=0}^{n-1} \mathbb{E}(\tilde{Z}_i)}. \quad (2.98)$$

This is a heuristic substitution and not an identity; the direction of the error can be pinned down exactly. The map $N \mapsto c^N$ is convex, so Jensen's inequality gives $\mathbb{E}(c^{\tilde{Z}_i}) \geq c^{\mathbb{E}(\tilde{Z}_i)}$: replacing the random census by its mean systematically understates the chance of getting through a step. With $m = 2p$,

$$\mathbb{E}(H_n) = \sum_{i=0}^{n-1} m^i = \begin{cases} \frac{1 - m^n}{1 - m}, & m \neq 1, \\ n, & m = 1, \end{cases} \quad (2.99)$$

and (2.98) is better read as the lower bound $\mathbb{P}([K = n] | \text{no catastrophe before } n) \geq c^{\mathbb{E}(H_n)}$ than as an equality. Nor are the steps independent, since surviving one is evidence of a smaller population, which raises the chance of surviving the next; both effects push the same way. How tight a bound remains a question for numerical work.

Non-rupture should not be confused with population survival. If the requirement is that the population be both unruptured and non-zero at generation n , the exact probability is

$$\mathbb{E}\left(\mathbf{1}_{\{\tilde{Z}_n > 0\}} c^{H_n}\right), \quad (2.100)$$

for which the Jensen bound does not directly provide an estimate. The distinction removes an ambiguity in the elementary approximation and indicates why the joint quasi-stationary problem is better posed through the killed-chain generator developed below.

The process can nevertheless be described exactly by a weighted generating function. Define

$$F_n(s) = \mathbb{E}\left(s^{\tilde{Z}_n} c^{H_n}\right), \quad F_0(s) = s. \quad (2.101)$$

Conditioning on the founder's reproduction — death with probability $1 - p$, or division with probability p , after which two independent subtrees each contribute a factor F_n and the founder's own step contributes c — gives the closed recursion

$$F_{n+1}(s) = c \phi(F_n(s)) = c \left(1 - p + p F_n(s)^2\right). \quad (2.102)$$

Its evaluations at special points have distinct meanings: $F_n(1)$ is the probability of no catastrophe in the first n updates, agreeing with (2.97), while $F_n(1) - F_n(0)$ is the probability of being both unruptured and non-extinct at generation n , the quantity of (2.100). The recursion (2.102) is exact, and it is the natural object to iterate when the Jensen bound is too crude.

The killed-chain viewpoint

The continuous-time analysis of birth–death–catastrophe processes proceeds through the generator restricted to the living states, and it is useful to fix the language here. Allow general state-dependent rates for the moment: from state $i \geq 1$, birth at rate λ_i , ordinary death at rate μ_i , and catastrophe at rate κ_i , with both absorbing outcomes lumped into a single cemetery state. On bounded test functions the generator is

$$(\mathcal{L}f)(i) = \lambda_i \{f(i+1) - f(i)\} + \mu_i \{f(i-1) - f(i)\} + \kappa_i \{f(0) - f(i)\}, \quad i \geq 1,$$

with $(\mathcal{L}f)(0) = 0$. Linear birth–death with per-capita catastrophe corresponds to $\lambda_i = i\lambda$, $\mu_i = i\mu$, $\kappa_i = i\rho$, recovering definition 2.2 once the cemetery is split into extinction and rupture.

Restricting $\mathcal{L}$ to the positive states gives the *killed subgenerator* K , with

$$K_{i,i+1} = \lambda_i, \quad K_{i,i-1} = \mu_i \quad (i \geq 2), \quad K_{i,i} = -(\lambda_i + \mu_i + \kappa_i).$$

The missing row mass is exactly the instantaneous absorption rate: κ_i for $i \geq 2$, and $\mu_1 + \kappa_1$ for $i = 1$, since from a single individual ordinary death is also absorption. Writing $T_0 = \inf\{t \geq 0 : X_t = 0\}$ and summing the positive-state forward equations gives the survival-loss balance

$$\frac{d}{dt} \mathbb{P}(T_0 > t) = -\mu_1 \mathbb{P}(X_t = 1) - \sum_{i \geq 1} \kappa_i \mathbb{P}(X_t = i), \quad (2.103)$$

where ordinary death removes survival mass only from state 1, while catastrophe can remove it from every positive state.

The balance sharpens the connection with quasi-stationarity. If a distribution $\nu = (\nu_i)_{i \geq 1}$ on the positive states is quasi-stationary and the relevant sums are finite, then for some $\theta > 0$,

$$\nu K = -\theta \nu, \quad \mathbb{P}([]_\nu T_0 > t) = e^{-\theta t}. \quad (2.104)$$

Indeed, quasi-stationarity writes the unnormalised positive-state law as $a(t)\nu$; the semigroup property forces $a(s+t) = a(s)a(t)$, and right-continuity gives $a(t) = e^{-\theta t}$; differentiating at zero yields the left-eigenvector relation. Summing its coordinates and comparing with (2.103) identifies the decay rate as

$$\theta = \mu_1 \nu_1 + \sum_{i \geq 1} \kappa_i \nu_i. \quad (2.105)$$

One limiting case is worth isolating. If the catastrophe clock is constant, $\kappa_i \equiv \kappa$ on the positive states, it may be taken independent of the underlying birth–death process, and the killed semigroup factorises as

$$P_t^K = e^{-\kappa t} P_t^{\text{BD}, K}.$$

The extra clock shifts the survival decay rate by κ but leaves the law conditional on survival unchanged. A state-dependent sequence such as $\kappa_i = i\rho$, by contrast, weights paths according to their accumulated population exposure and generally changes the quasi-stationary law itself. Rates, means and variances, state probabilities and the associated generating-function equations for [definition 2.2](#) are developed in the later chapters on birth–death–catastrophe processes, with the two-type extension treated in the chapter on multi-type processes and the release of a ruptured compartment’s contents into a shared medium in the chapter on compartment rupture.

2.G Further absorption models

The models considered so far describe the population inside a single compartment. Once several compartments share a medium, a further question arises: what becomes of the contents of the first compartment to rupture? The shared medium acquires, at a stroke, a potentially large cohort of particles available for subsequent absorption by the compartments that remain. This appendix treats the case in which a cohort of particles enters a single compartment, by the method of [section 2.3.3](#): the absorption-only model, which is simpler than the worked example of [section 2.3.3.2](#) and serves as a check case, and the absorption–birth–death model, which is harder than it and whose closed form, [proposition 2.6](#), is a result of this thesis.

2.G.1 Absorption only

The simplest absorption process has a single compartment that takes in particles, with no subsequent dynamics: no birth and no death within the compartment.

Let X_t and Y_t count the particles inside and outside the compartment respectively; the same labelling is used for every version of the single-compartment model below. Events are stochastic with exponentially distributed inter-event times, which by [section 2.2.2.1](#) follows from the assumption that event rates do not depend on time, so that $\{(X_t, Y_t) : t \geq 0\}$ is a continuous-time Markov chain in the sense of [section 2.2.2.2](#).

With only absorption to consider, the model is specified by the transition probabilities over an infinitesimal interval Δt ,

$$\mathbb{P}(\Delta X = 1, \Delta Y = -1 \mid Y_t = j) = j\alpha \Delta t, \quad (\text{absorption})$$

$$\mathbb{P}(\Delta X = 0, \Delta Y = 0 \mid Y_t = j) = 1 - j\alpha \Delta t, \quad (\text{no event})$$

where $\Delta X = X_{t+\Delta t} - X_t$ and similarly for Y , and terms of $o(\Delta t)$ are suppressed throughout. The factor j records that each exterior particle is independently subject to the same probability $\alpha \Delta t$ of being taken up. The initial condition is $X_0 = 0, Y_0 = N$: a starter population of N particles occupying the exterior medium.

Writing $p_{i,j}(t) = \mathbb{P}(X_t = i, Y_t = j)$, the forward Kolmogorov equations are

$$\frac{dp_{i,j}}{dt} = \alpha(j+1)p_{i-1,j+1} - \alpha j p_{i,j}. \quad (2.106)$$

With the bivariate generating function

$$G(x, y, t) = \sum_i \sum_j p_{i,j}(t) x^i y^j = \mathbb{E}(x^{X_t} y^{Y_t}), \quad (2.107)$$

(2.106) converts into a partial differential equation by the procedure of [section 2.3.1.1](#). Multiply each equation by $x^i y^j$ and sum over i and j ; the sums are finite, because $p_{i,j}(t) = 0$ whenever $i + j \neq N$, particles in this model being neither created nor destroyed. Shifting indices in the first term sends $\alpha(j+1)p_{i-1,j+1}x^i y^j$ to $\alpha x \partial_y G$, and the second term is $-\alpha y \partial_y G$, so that

$$\frac{\partial G}{\partial t} = \alpha(x - y) \frac{\partial G}{\partial y}, \quad G(x, y, 0) = y^N. \quad (2.108)$$

The method of characteristics of [section 2.3.3.1](#) applies to (2.108) as it does to (2.58), but the solution can be obtained more quickly by exploiting the independence that this model retains, and that route also provides a check on the machinery.

Exploiting independence

The function G above was specific to $X_0 = 0, Y_0 = N$. Define more generally

$$G_n(x, y, t) = \sum_i \sum_j \mathbb{P}(X_t = i, Y_t = j \mid X_0 = 0, Y_0 = n) x^i y^j.$$

In a process of pure absorption the n particles are taken up one by one, and the behaviour of any one of them does not depend on the movements of its neighbours. That independence lets X_t and Y_t decompose into sums of n single-particle variables,

$$X_t = \sum_{k=1}^n X_t^{(k)}, \quad Y_t = \sum_{k=1}^n Y_t^{(k)},$$

from which

$$G_n(x, y, t) = (G_1(x, y, t))^n, \quad (2.109)$$

since

$$G_n(x, y, t) = \mathbb{E}\left(\prod_{k=1}^n x^{X_t^{(k)}} \prod_{k=1}^n y^{Y_t^{(k)}}\right) = \prod_{k=1}^n \mathbb{E}\left(x^{X_t^{(k)}} y^{Y_t^{(k)}}\right) = (G_1(x, y, t))^n.$$

Now consider a single initial particle, $X_0 = 0, Y_0 = 1$. It is eventually absorbed, and that is the final state of the system: no other states are reachable, so $p_{i,j}(t) = 0$ whenever $i + j \neq 1$. The forward equations reduce to

$$\frac{dp_{0,1}}{dt} = -\alpha p_{0,1}, \quad \frac{dp_{1,0}}{dt} = \alpha p_{0,1},$$

giving $p_{0,1}(t) = e^{-\alpha t}$ and $p_{1,0}(t) = 1 - e^{-\alpha t}$ (figure 2.16). These being the only two non-zero state probabilities of the single-particle system,

$$G_1(x, y, t) = p_{1,0}(t)x + p_{0,1}(t)y = (1 - e^{-\alpha t})x + e^{-\alpha t}y,$$

and (2.109) gives the generating function for any initial census,

$$G_n(x, y, t) = \left((1 - e^{-\alpha t})x + e^{-\alpha t}y \right)^n. \quad (2.110)$$

This satisfies (2.108) together with the initial condition $G_n(x, y, 0) = y^n$. The form is recognisable: $X_t \sim \text{Binomial}(n, 1 - e^{-\alpha t})$, as it must be, since the n particles are absorbed independently and each has been taken up by time t with probability $1 - e^{-\alpha t}$.

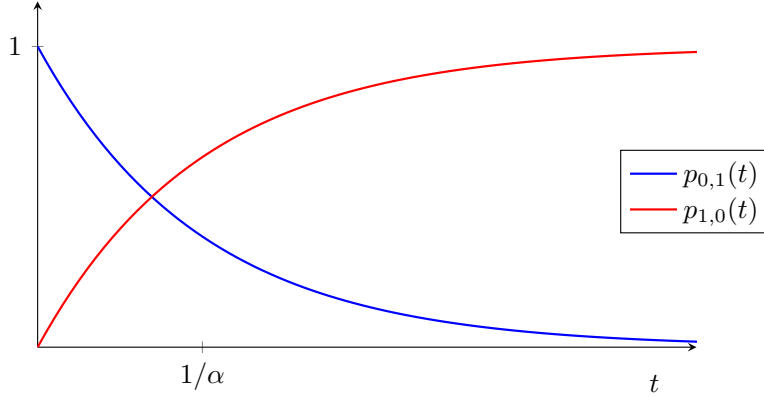


Figure 2.16: State probabilities for the single-particle absorption-only model. The particle begins outside the compartment ($p_{0,1}$, decaying exponentially at rate α) and ends inside it ($p_{1,0}$). There is nowhere else for it to go, so the two curves sum to one at every time.

2.G.2 Absorption–birth–death

Introducing replication for particles inside the compartment, the independence property is still present but no longer serves as a shortcut: from a single initial particle, birth events make infinitely many states reachable, so $G_1(x, y, t)$ is an infinite sum and there is nothing to be gained by factorising. The partial differential equation must therefore be solved directly, and by now that can be done.¹

With X_t and Y_t as before, birth at per-capita rate λ is added to the transition probabilities:

$$\begin{aligned} \mathbb{P}(\Delta X = 1, \Delta Y = -1 \mid Y_t = j) &= j\alpha \Delta t, & (\text{absorption}) \\ \mathbb{P}(\Delta X = 1, \Delta Y = 0 \mid X_t = i) &= i\lambda \Delta t, & (\text{birth}) \\ \mathbb{P}(\Delta X = -1, \Delta Y = 0 \mid X_t = i) &= i\mu \Delta t, & (\text{death}) \\ \mathbb{P}(\Delta X = 0, \Delta Y = 0 \mid X_t = i, Y_t = j) &= 1 - (j\alpha + i\mu + i\lambda)\Delta t, & (\text{no event}) \end{aligned}$$

with forward Kolmogorov equation

$$\frac{dp_{i,j}}{dt} = \alpha(j+1)p_{i-1,j+1} + \mu(i+1)p_{i+1,j} + \lambda(i-1)p_{i-1,j} - (\alpha j + \mu i + \lambda i)p_{i,j}. \quad (2.111)$$

¹I am grateful to Benjamin Sharp for the advice on applying the method of characteristics to three-variate problems that unlocked this calculation.

The same reasoning as before gives the generating-function equation

$$\frac{\partial G}{\partial t} = (\mu - (\mu + \lambda)x + \lambda x^2) \frac{\partial G}{\partial x} + \alpha(x - y) \frac{\partial G}{\partial y}, \quad G(x, y, 0) = y^N, \quad (2.112)$$

whose x -coefficient is exactly the quadratic met in (2.18) for the linear birth–death process. Reparametrising $(x, y, t) \rightarrow (\sigma, \eta, \tau)$ as in the worked example of section 2.3.3.2 gives the characteristic system

$$\frac{dG}{d\tau} = 0, \quad (2.113)$$

$$\frac{dt}{d\tau} = -1, \quad (2.114)$$

$$\frac{dx}{d\tau} = \mu - (\mu + \lambda)x + \lambda x^2, \quad (2.115)$$

$$\frac{dy}{d\tau} = \alpha(x - y), \quad (2.116)$$

with the same initial conditions (2.56).

The equations for G and t , (2.113)–(2.114). As before,

$$G(\sigma, \eta, \tau) = \eta^N, \quad t = -\tau. \quad (2.117)$$

The equation for x , (2.115). This is a Riccati equation of the same shape as the one governing the survival probability in section 2.C. Its right-hand side factorises, and the roots are elementary rather than the surd expression that mechanical use of the quadratic formula suggests:

$$\mu - (\mu + \lambda)x + \lambda x^2 = \lambda(x - x_+)(x - x_-), \quad x_+ = 1, \quad x_- = \frac{\mu}{\lambda}. \quad (2.118)$$

The discriminant is $(\frac{\mu+\lambda}{2\lambda})^2 - \frac{\mu}{\lambda} = (\frac{\lambda-\mu}{2\lambda})^2$, a perfect square, so the surds cancel. It is convenient to name the combination that appears throughout,

$$b_1 = \lambda(x_+ - x_-) = \lambda - \mu, \quad (2.119)$$

the net interior growth rate. Separating variables in (2.115) and applying (2.56) gives the characteristic in the compact form

$$\frac{x - x_+}{x - x_-} = \left(\frac{\sigma - x_+}{\sigma - x_-} \right) e^{b_1 \tau}, \quad (2.120)$$

or, solved for x ,

$$x(\tau) = \frac{x_+(\sigma - x_-) - x_-(\sigma - x_+)e^{b_1 \tau}}{(\sigma - x_-) - (\sigma - x_+)e^{b_1 \tau}}. \quad (2.121)$$

The equation for y , (2.116). The last characteristic equation is $dy/d\tau = \alpha(x - y)$, which with an integrating factor becomes

$$\frac{d}{d\tau} (e^{\alpha \tau} y) = \alpha e^{\alpha \tau} x(\tau). \quad (2.122)$$

The integral on the right is where the difficulty lies. Rather than substitute (2.121) and split the resulting quotient — which leads to two hypergeometric integrals and a good deal of bookkeeping —

it is cleaner to expand $x(\tau)$ first. Writing $x - x_- = (x - x_+) + (x_+ - x_-)$ in (2.120) and solving for x gives

$$x(\tau) = x_+ + (x_+ - x_-) \frac{\zeta(\tau)}{1 - \zeta(\tau)}, \quad \zeta(\tau) = \left(\frac{\sigma - x_+}{\sigma - x_-} \right) e^{b_1 \tau}, \quad (2.123)$$

so that, for $|\zeta| < 1$,

$$x(\tau) = x_+ + (x_+ - x_-) \sum_{n=1}^{\infty} \zeta(\tau)^n.$$

Every term is now a pure exponential in τ and integrates immediately. Writing $Z = (\sigma - x_+)/(\sigma - x_-)$, so that $\zeta(\tau) = Ze^{b_1 \tau}$ and $Z^n e^{nb_1 \tau} = \zeta^n$,

$$\begin{aligned} \alpha \int e^{\alpha \tau} x(\tau) d\tau &= \alpha x_+ \int e^{\alpha \tau} d\tau + \alpha(x_+ - x_-) \sum_{n=1}^{\infty} Z^n \int e^{(\alpha + nb_1)\tau} d\tau \\ &= e^{\alpha \tau} \left[x_+ + \alpha(x_+ - x_-) \sum_{n=1}^{\infty} \frac{\zeta^n}{\alpha + nb_1} \right] + \text{const.} \end{aligned}$$

Setting $\delta = \alpha/b_1$ and using $\alpha(x_+ - x_-)/b_1 = \alpha/\lambda$ by (2.119), this defines the function that carries the whole solution:

$$\Psi(\zeta) = x_+ + \frac{\alpha}{\lambda} \sum_{n=1}^{\infty} \frac{\zeta^n}{n + \delta}, \quad \delta = \frac{\alpha}{b_1} = \frac{\alpha}{\lambda - \mu}. \quad (2.124)$$

So $\alpha \int e^{\alpha \tau} x(\tau) d\tau = e^{\alpha \tau} \Psi(\zeta(\tau)) + \text{const}$, and integrating (2.122) gives $e^{\alpha \tau} y = e^{\alpha \tau} \Psi(\zeta(\tau)) + \text{const}$. The initial condition $y(\sigma, \eta, 0) = \eta$, at which $\zeta(0) = Z$, fixes the constant as $\eta - \Psi(Z)$, so

$$\eta = e^{\alpha \tau} \left[y - \Psi(\zeta(\tau)) \right] + \Psi(Z). \quad (2.125)$$

Finally, put $\tau = -t$ and use (2.120): writing

$$W = \frac{x - x_+}{x - x_-}$$

we have $\zeta(\tau) = W$ and $Z = We^{-b_1 \tau} = We^{b_1 t}$. Substituting into (2.125) and applying (2.117) gives the solution.

Proposition 2.6 (Generating function for absorption–birth–death). *With $x_+ = 1$, $x_- = \mu/\lambda$, $b_1 = \lambda - \mu$, $\delta = \alpha/b_1$, $W = (x - x_+)/(\sigma - x_-)$ and Ψ as in (2.124),*

$$\boxed{G(x, y, t) = \left\{ e^{-\alpha t} [y - \Psi(W)] + \Psi(We^{b_1 t}) \right\}^N}. \quad (2.126)$$

At resonant parameter values, described in remark 2.8, the right-hand side is understood by its finite limit.

Three checks confirm the result. At $t = 0$ the two Ψ terms cancel and $G = y^N$, the required initial condition. At $x = y = 1$ we have $W = 0$ and $\Psi(0) = x_+ = 1$, so $G = 1$ and probability is conserved. And setting $x = 1$ alone gives

$$G(1, y, t) = (e^{-\alpha t} y + 1 - e^{-\alpha t})^N,$$

so $Y_t \sim \text{Binomial}(N, e^{-\alpha t})$ — exactly right, since the exterior particles are absorbed independently at rate α and what happens inside the compartment cannot affect them. Beyond these, (2.126) has been verified numerically against direct integration of the master equation (2.111), agreeing to around 10^{-13} across sub- and supercritical interior regimes and for $N = 1, 2, 3$.

2.G.2.1 A parameter-regular representation

The special-function form is compact, but the underlying probability calculation provides a second representation that remains transparent at $\lambda = \mu$, at resonance and across hypergeometric branch cuts. Let $F(x, t)$ be the probability generating function at time t of an interior linear birth–death process started from one particle. Its backward equation is

$$\frac{\partial F}{\partial t} = \lambda F^2 - (\lambda + \mu)F + \mu, \quad F(x, 0) = x. \quad (2.127)$$

For $\lambda \neq \mu$,

$$\frac{F(x, t) - 1}{F(x, t) - x_-} = \frac{x - 1}{x - x_-} e^{b_1 t}, \quad (2.128)$$

while at $\lambda = \mu$,

$$F(x, t) = 1 - \frac{1 - x}{1 + \lambda t(1 - x)}. \quad (2.129)$$

For one initial exterior particle, either transfer has not occurred by time t , or it occurs at some time $u \in (0, t)$ and leaves an interior birth–death process of duration $t - u$. Conditioning on the transfer time gives

$$g(x, y, t) = e^{-\alpha t} y + \alpha e^{-\alpha t} \int_0^t e^{\alpha v} F(x, v) dv, \quad (2.130)$$

and independence of the N initial particles gives

$$G(x, y, t) = g(x, y, t)^N. \quad (2.131)$$

This formula is real and finite for $x, y \in [0, 1]$ and all positive rates. For $\lambda \neq \mu$, substituting (2.128) into the integral and using the antiderivative behind (2.124) recovers (2.126); thus (2.130) also specifies unambiguously which continuation of the special-function expression is intended wherever the series representation breaks down.

Several checks now require almost no algebra. At $t = 0$, (2.130) gives $g = y$, and hence $G = y^N$. At $x = y = 1$, $F(1, v) = 1$, so $g = 1$ and total probability is conserved. Setting $x = 1$ gives

$$G(1, y, t) = (e^{-\alpha t} y + 1 - e^{-\alpha t})^N, \quad (2.132)$$

which is the generating function of $Y_t \sim \text{Binomial}(N, e^{-\alpha t})$: the exterior marginal is unaffected by the interior birth and death rates, as the transition diagram requires.

2.G.2.2 Hypergeometric form, domains and resonance

The series in (2.124) is a Gauss hypergeometric function in disguise. With

$${}_2F_1(a, b; c; z) = \sum_{n=0}^{\infty} \frac{(a)_n (b)_n}{(c)_n} \frac{z^n}{n!}, \quad (q)_n = \begin{cases} 1, & n = 0, \\ q(q+1) \cdots (q+n-1), & n \geq 1, \end{cases} \quad (2.133)$$

where $(q)_n$ is the rising Pochhammer symbol [10], the case $a = 1$, $b = \delta$, $c = \delta + 1$ collapses to a single ratio. Since $(1)_n = n!$ and

$$\frac{(\delta)_n}{(\delta+1)_n} = \frac{\delta(\delta+1) \cdots (\delta+n-1)}{(\delta+1) \cdots (\delta+n-1)(\delta+n)} = \frac{\delta}{\delta+n}, \quad (2.134)$$

we have

$${}_2F_1(1, \delta; \delta + 1; z) = \sum_{n=0}^{\infty} \frac{\delta}{\delta + n} z^n. \quad (2.135)$$

Comparing with (2.124) and using $\alpha/(\lambda\delta) = b_1/\lambda = x_+ - x_-$,

$$\Psi(\zeta) = x_+ + (x_+ - x_-) \left[{}_2F_1(1, \delta; \delta + 1; \zeta) - 1 \right]. \quad (2.136)$$

(2.126) together with (2.136) is the closed form. It is not pretty, but it is a genuine closed form in terms of a standard special function, and the elementary series (2.124) is what one would actually evaluate. An alternative derivation of the same object, by way of the integral $\int e^{h\tau}/(P + Qe^{k\tau}) d\tau$, is given in section 2.A; that route is longer, but it produces a general identity of independent use.

Remark 2.7 (Domain of validity). The series (2.124) converges for $|\zeta| < 1$, and the arguments appearing in (2.126) are W and $We^{b_1 t}$. In the subcritical interior regime $\mu > \lambda$ we have $b_1 < 0$ and $x_- = \mu/\lambda > 1$, so for $x \in [0, 1]$ the argument W lies in $[0, \lambda/\mu] \subset [0, 1)$ and $We^{b_1 t}$ is smaller still: the series representation is valid on the whole of the unit square, which is the case of most interest here. When $\lambda > \mu$ the arguments may leave the unit disc or cross the standard hypergeometric cut, and one must use the analytic continuation of ${}_2F_1$, tracking the branch along the characteristic — conveniently, the Euler integral

$${}_2F_1(1, \delta; \delta + 1; \zeta) = \delta \int_0^1 \frac{u^{\delta-1}}{1 - \zeta u} du \quad (\delta > 0, \zeta \notin [1, \infty)),$$

which was used for the numerical verification quoted above, or the real integral (2.130), which is usually safer still. The apparent singularity of W at $x = x_-$ is only a coordinate singularity: $F(x_-, t) = x_-$, and the continuous limit of (2.130) is regular there.

Remark 2.8 (Resonance). The denominators in (2.124) vanish when $n + \delta = 0$ for some integer $n \geq 1$, that is when

$$\alpha = n(\mu - \lambda), \quad n = 1, 2, \dots$$

This is the classical resonance condition for linearising the y -equation: the decay rate α of the absorption channel coincides with a harmonic of the interior relaxation rate $\mu - \lambda$. The generating function itself is perfectly regular there, being analytic in all parameters, so the singularity is removable. In the resonant term, the combination

$$\frac{(We^{b_1 t})^n - e^{-\alpha t} W^n}{n + \delta}$$

that appears when the two Ψ evaluations in (2.126) are differenced has a finite parameter limit and produces the familiar secular factor $t e^{-\alpha t}$ in place of the simple exponential. Equivalently, the convolution (2.130) contains no singular denominator and supplies the limit directly. The resonant parameter values form a set of measure zero and the limit may be taken numerically, but the degeneracy should be kept in mind when α happens to sit near an integer multiple of $\mu - \lambda$.

2.G.3 Recovering state probabilities

A generating function is useful only insofar as its coefficients can be recovered. In every model of this appendix,

$$p_{i,j}(t) = [x^i y^j] G(x, y, t) = \frac{1}{i! j!} \partial_x^i \partial_y^j G \Big|_{x=y=0}, \quad (2.137)$$

the bivariate Maclaurin coefficient of [section 2.B](#), which may alternatively be extracted by Cauchy's formula on circles inside the domain of analyticity. For absorption–birth–death the dependence on y remains linear before the power N is taken: writing

$$h(x, t) = \alpha e^{-\alpha t} \int_0^t e^{\alpha v} F(x, v) dv, \quad g(x, y, t) = e^{-\alpha t} y + h(x, t)$$

as in [\(2.130\)](#), the binomial theorem separates the two coefficient extractions,

$$p_{i,j}(t) = \binom{N}{j} e^{-\alpha t j} [x^i] h(x, t)^{N-j}, \quad 0 \leq j \leq N, \quad (2.138)$$

and the remaining univariate coefficient can be obtained by differentiation, by convolution of truncated series, or by Cauchy's integral formula. If $h(x, t) = \sum_{k \geq 0} h_k(t) x^k$, the convolution is

$$[x^i] h(x, t)^M = \sum_{\substack{k_1 + \dots + k_M = i \\ k_r \geq 0}} \prod_{r=1}^M h_{k_r}(t),$$

in which truncating h after degree i is exact: coefficients of higher degree cannot contribute to $[x^i]$. As a check, summing over all interior states at fixed j means setting $x = 1$; since $h(1, t) = 1 - e^{-\alpha t}$,

$$\sum_{i=0}^{\infty} p_{i,j}(t) = \binom{N}{j} e^{-\alpha t j} (1 - e^{-\alpha t})^{N-j},$$

the exterior binomial marginal of [\(2.132\)](#).

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